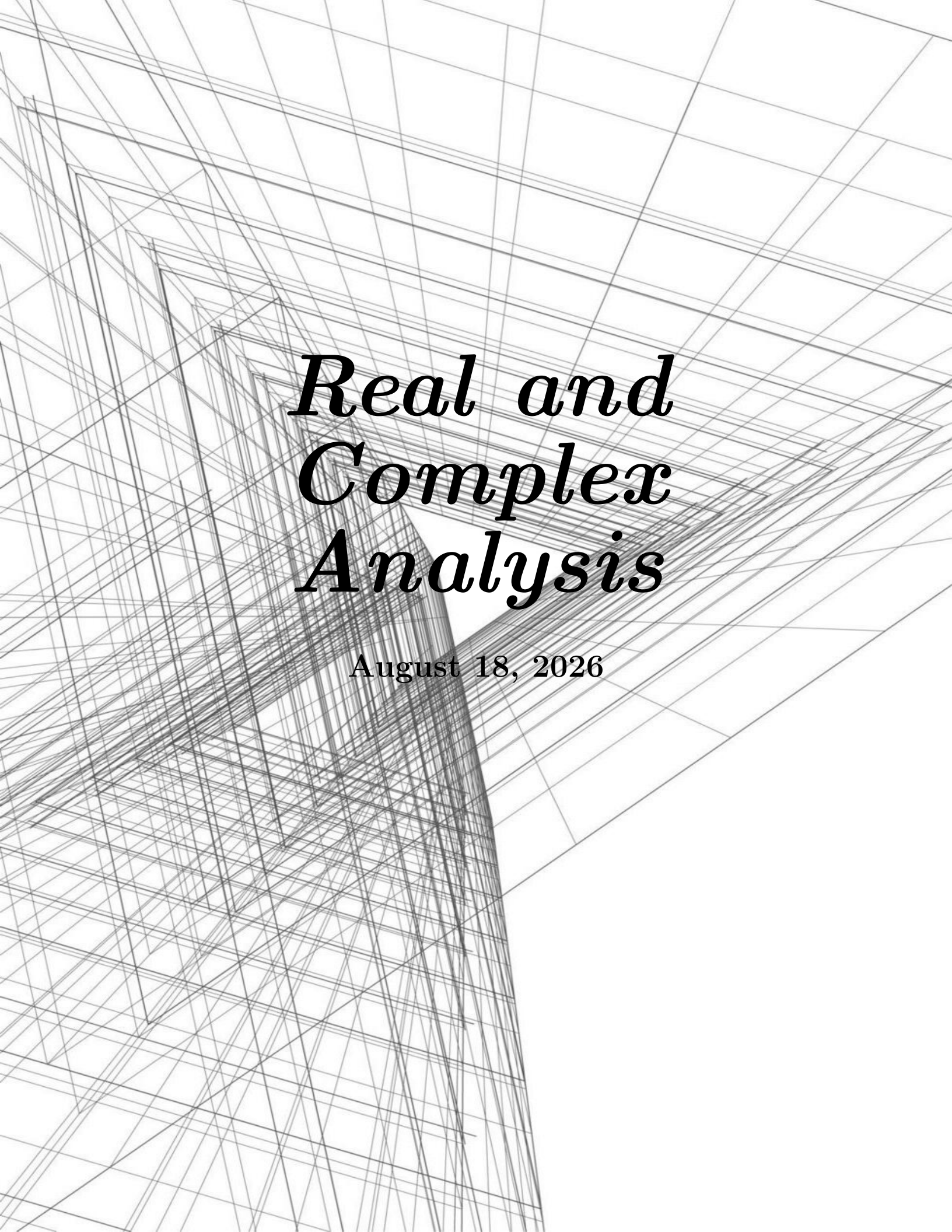




Real and Complex Analysis

August 18, 2026

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Chapter 1

The Real Numbers

We assume that the set of real numbers, which is denoted by $\mathbb{R}$ satisfies three types of axioms.

NOTICE: IN THIS BOOK WE DENOTE THE NATURAL NUMBERS $\mathbb{N} = \mathbb{Z}_+$ WITHOUT 0.

1.1 The Axioms

The axioms are as follows:

The Field Axioms $\mathbb{R}$ is a field: there are binary operation $+, \cdot$ on $\mathbb{R}$ that:

- Commutativity of Addition: $\forall a, b \in \mathbb{R}, a + b = b + a$.
- Associativity of Addition: $\forall a, b, c \in \mathbb{R}, a + (b + c) = (a + b) + c$.
- The Additive Identity: $\exists 0 \in \mathbb{R}, \forall a \in \mathbb{R}, a + 0 = 0 + a = a$.
- The Additive Inverse: $\forall a \in \mathbb{R}, \exists b \in \mathbb{R}, a + b = 0$.
- Commutativity of Multiplication: $\forall a, b \in \mathbb{R}, ab = ba$.
- Associativity of Multiplication: $\forall a, b, c \in \mathbb{R}, (ab)c = a(bc)$.
- The Multiplicative Identity: $\exists 1 \in \mathbb{R}, \forall a \in \mathbb{R}, 1a = a1 = a$.
- The Multiplicative Inverse: $\forall a \neq 0 \in \mathbb{R}, \exists b \in \mathbb{R}, ab = 1$.
- The Distributive Property: $\forall a, b, c \in \mathbb{R}, a(b + c) = ab + ac$.
- The Nontriviality Assumption: $1 \neq 0$.

Any set that satisfies these axioms is called a **field**. It follows from the commutativity of addition that the additive identity, 0, is unique, and we infer from the commutativity of multiplication that the multiplicative unit, 1, also is unique. The additive inverse and multiplicative inverse also are unique, denoted by $-a$ and a^{-1} respectively.

The Positivity Axioms In the real numbers there is a natural notion of order: greater than, less than, and so on. A convenient way to codify these properties is by specifying axioms satisfied by the set of positive numbers.

There is a set of real numbers, denoted by $\mathcal{P}$, called the set of **positive numbers**, having the following properties:

- If $a, b \in \mathcal{P}$, then $ab, a + b \in \mathcal{P}$.
- $\forall a \in \mathbb{R}$ exactly one of the following three alternatives is true: $a \in \mathcal{P}, a = 0, -a \in \mathcal{P}$.

The Positivity Axioms lead in a natural way to an ordering of the real numbers: for real numbers a, b , we define $a > b$ to mean $a - b$ is positive, and other ordering relation is defined similarly.

Using the Field Axioms and the Positivity Axioms, it is possible to formally establish the familiar properties of inequalities. We define the intervals:

$$\begin{aligned} (a, b) &= \{x : a < x < b\}, [a, b] = \{x : a \leq x \leq b\} \\ [a, b) &= \{x : a \leq x < b\}, (a, b] = \{x : a < x \leq b\}. \end{aligned} \quad (1.1)$$

The Completeness Axiom A nonempty subset $E \subseteq \mathbb{R}$ is called **bounded above** if $\exists b \in \mathbb{R}, \forall x \in E, x \leq b$. The number b is called the upper bound for E . A lower bound is defined similarly.

The completeness axiom is as follows

Let E be a nonempty subset of $\mathbb{R}$ that is bounded above. Then the set of upper bounds for E has a minimum value, called the least upper bound, or **supremum**, denoted by $\sup E$.

Similarly, the greatest lower bound is called the **infimum** denoted by $\inf E$.

The Triangle Inequality We define the absolute value of a real number x , to be x if $x \geq 0$ and $-x$ if $x < 0$, denoted by $|x|$.

The following inequality, called the Triangle Inequality, is fundamental in mathematical analysis:

$$\forall a, b \in \mathbb{R}, |a + b| \leq |a| + |b|. \quad (1.2)$$

The Extended Real Numbers It is convenient to introduce the symbol ∞ and $-\infty$ and write

$$\forall x \in \mathbb{R}, -\infty < x < \infty \quad (1.3)$$

we called $\mathbb{R} \cup \pm\infty$ the **extended real numbers**. If a nonempty $E \subseteq \mathbb{R}$ has no upper bound we denote $\sup E = \infty$, and if $E = \emptyset$ we denote $\sup E = -\infty$, (which is perfectly true because every real number is an upper bound for an empty set.) Therefore every set of real numbers has a supremum that belongs to the extended real-numbers. Similarly, we can extend the concept of infimum so every set of real numbers has an infimum that belongs to the extended real numbers. We will define limits of sequences of real numbers and it is convenient to allow limits that are extended real numbers. Many properties of sequences of real numbers that converge to real numbers, such as the limit of the sum is the sum of the limits and the limit of the product is the product of the limit, continue to hold for limits that are $\pm\infty$, provided we make the following extension of the meaning of sum and product:

- $\infty + \infty = \infty, -\infty - \infty = -\infty.$
- $\forall x \in \mathbb{R}, x + \infty = \infty, x - \infty = -\infty.$
- $\forall x \in \mathbb{R}, x > 0$ we have $x \cdot \infty = \infty, x \cdot (-\infty) = -\infty.$
- $\forall x \in \mathbb{R} < x < 0$ we have $x \cdot \infty = -\infty, x \cdot (-\infty) = \infty.$

We also define

$$\begin{aligned} (a, \infty) &= \{x \in \mathbb{R} : a < x\}, (-\infty, b) = \{x \in \mathbb{R} : x < b\} \\ [a, \infty) &= \{x \in \mathbb{R} : a \leq x\}, (-\infty, b] = \{x \in \mathbb{R} : x \leq b\} \end{aligned} \quad (1.4)$$

Sets of the above form are unbounded intervals.

1.2 Countable and Uncountable Sets

We call two sets A, B equipotent if there is a one-to-one mapping $f : A \rightarrow B$. Equipotence defines an equivalence relation among sets, that is, it is reflexive, symmetric, and transitive.

We denote $\{1, 2, \dots, n\} = \{k \in \mathbb{Z}_+ : 1 \leq k \leq n\}$. The first observation regarding equipotence is that for any positive integers n, m , the set $\{1, \dots, n + m\}$ is not equipotent to $\{1, \dots, n\}$. This observation is often called the **pigeonhole principle** and may be proved by an induction argument with respect to n .

Definition 1.2.1: Sets Classification

A set E is called finite if it is empty or there is a positive integer n such that E is equipotent to $\{1, \dots, n\}$.

We call E **countably infinite** if E is equipotent to $\mathbb{Z}_+$. A set that is either finite or countably infinite is called **countable** and otherwise **uncountable**.

Proposition: Subsets Countability

A subset of a countable set is countable. In particular, every set of natural numbers is countable.

Example: Countable Sets

For all $n \in \mathbb{N}$, we have $\mathbb{N}^n$ and $\mathbb{Q}$ are countably infinite.

Proposition: Image Countability

A nonempty set is countable if and only if it is the image of a function whose domain is a nonempty countable set.

Proposition:

The union of a countable collection of countable sets is countable.

Theorem 1.2.1: Real Number Countability

A nondegenerate interval of real numbers is uncountable.

Proof. Let I be a nondegenerate interval of $\mathbb{R}$. Clearly I is not finite.

Suppose I is countable, let $\{x_n : n \in \mathbb{N}\}$ be an enumeration of I . Let $[a_1, b_1]$ be a nondegenerate closed, bounded subinterval of I which does not contain x_1 . Then let $[a_2, b_2] \subseteq [a_1, b_1]$ be a nondegenerate subset does not contain x_2 , and go on. For each n , we have $x_n \notin [a_n, b_n]$. So we have an ascending sequence a_n and descending b_n . The completeness axiom tell us that there is $x^* = \sup \{a_n\}$. As all b_n are upper bounds of a_n so we have $x^* \leq b_n$. We found an x^* that is not in any x_n . $\square$

1.3 Open, Closed and Borel Sets

The natural topology on $\mathbb{R}$ is given by the metric $d(x, y) = |x - y|$. This induces open and closed sets.

Definition 1.3.1: Open Sets

A set $\mathcal{O} \subseteq \mathbb{R}$ is open if $\forall x \in \mathcal{O}$, there is an interval $(x - r, x + r) \subseteq \mathcal{O}$.

Proposition: The Topology of Real Numbers

The set of real numbers $\mathbb{R}$ and the empty-set $\emptyset$ are open; the intersection of any finite collection of open sets is open; and the union of any collection of open sets is open.

Proposition: Open Set Composition Theorem

Every nonempty open set is the disjoint union of a countable collection of open intervals.

Proof. Let $\mathcal{O}$ be a nonempty open set. For each $x \in \mathcal{O}$, there is $y > x$ and $z < x$ such that $(z, y) \subseteq \mathcal{O}$. Let extended real numbers $a_x = \inf \{z : (z, y) \subseteq \mathcal{O}\}$ and $b_x = \sup \{y : (z, y) \subseteq \mathcal{O}\}$. Then $x \in (a_x, b_x)$. We claim that

$$I_x = (a_x, b_x) \subseteq \mathcal{O}, a_x \notin \mathcal{O}, b_x \notin \mathcal{O}.$$

which can be verified easily. Consider the collection of intervals $\{I_x : x \in \mathcal{O}\}$, we have

$$\mathcal{O} = \bigcup_{x \in \mathcal{O}} I_x.$$

and this collection is disjoint. (because endpoints of each interval are not in $\mathcal{O}$).

It remains to show that this collection is countable. By the density of the rationals, each of these open intervals contains a rational number. This establishes a one-to-one correspondence between the collection of open intervals and a subset of the rational numbers. As any set of rational numbers is countable, therefore $\mathcal{O}$ is the union of a countable disjoint collection of open intervals. $\square$

Theorem 1.3.1: The Heine-Borel Theorem

Every closed and bounded set of real numbers is compact, that is, every open cover of the set has a finite subcover.

Definition 1.3.2: Descending and Ascending Sets

We say that a sequence of sets $\{E_n\}$ is descending if $E_1 \supseteq E_2 \supseteq E_3 \supseteq \dots$ and ascending if $E_1 \subseteq E_2 \subseteq E_3 \subseteq \dots$.

Theorem 1.3.2: The Nested Set Theorem

Let $\{E_n\}$ be a descending sequence of nonempty closed and bounded sets of real numbers. Then the intersection $\bigcap_{n=1}^{\infty} E_n$ is nonempty.

This compactness leads to many general results, like the nested set theorem.

Definition 1.3.3: σ -algebra

Given a set X , a σ -algebra $\mathcal{A}$ is a collection of subsets of X such that:

- $\emptyset \in \mathcal{A}$.
- If $A \in \mathcal{A}$, then its complement $X \setminus A \in \mathcal{A}$.
- The union of a countable collection of sets in $\mathcal{A}$ is also in $\mathcal{A}$, that is, if $A_1, A_2, \dots \in \mathcal{A}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

Remark:

As a comparison, a topology $\mathcal{T}$ on a set X requires closure under arbitrary unions and finite intersections, while a σ -algebra requires closure under countable unions and (because of complements) countable intersections.

Given a set X , the collection $\{\emptyset, X\}$ is the smallest σ -algebra on X , and the power set 2^X is the largest σ -algebra on X . Also, as $\emptyset \in \mathcal{A}$, we infer that finite unions and finite intersections of sets in $\mathcal{A}$ are also in $\mathcal{A}$.

We also observe that a σ -algebra is closed under relative complements: if $A, B \in \mathcal{A}$, then $A \setminus B = A \cap (X \setminus B) \in \mathcal{A}$.

Theorem 1.3.3: Generating Minimal σ -algebra

Let $\mathcal{F}$ be a collection of subsets of X . Then the intersection of all σ -algebras containing $\mathcal{F}$ is the smallest σ -algebra containing $\mathcal{F}$, in the sense that it is contained in every σ -algebra containing $\mathcal{F}$.

We need to introduce the concept of limit superior and limit inferior of a sequence of sets.

Definition 1.3.4: Limit Superior and Limit Inferior of Sets

Let $\{E_n\}$ be a sequence of sets. The limit superior of the sequence is defined as

$$\limsup_{n \rightarrow \infty} E_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k, \quad (1.5)$$

and the limit inferior of the sequence is defined as

$$\liminf_{n \rightarrow \infty} E_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} E_k. \quad (1.6)$$

Remark:

Intuitively speaking, $\limsup_{n \rightarrow \infty} E_n$ consists of the elements that belong to infinitely many of the sets E_n , while $\liminf_{n \rightarrow \infty} E_n$ consists of the elements that belong to all but finitely many of the sets E_n .

Let's compare this to that of a sequence of real numbers $\{x_n\}$. The limit superior of the sequence is defined as

$$\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} x_k,$$

which means the largest limit point of the sequence, while the limit inferior of the sequence is defined as

$$\liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} x_k,$$

which means the smallest limit point of the sequence.

Not let $\{A_n\}$ be a sequence of sets that belong to a σ -algebra $\mathcal{A}$. Then we have

$$\limsup_{n \rightarrow \infty} A_n, \liminf_{n \rightarrow \infty} A_n \in \mathcal{A}. \quad (1.7)$$

because $\mathcal{A}$ is closed under countable unions and intersections.

Although the union of any collection of open sets is open and the intersection of any finite collection of open sets is open, as we have seen, the intersection of a countable collection of open sets need not be open. Here, we shall see that the smallest σ -algebra containing all open sets on the real line is quite worthy of attention.

Definition 1.3.5: The Borel σ -algebra

The smallest σ -algebra containing all open sets of $\mathbb{R}$ is called the **Borel σ -algebra** and is denoted by $\mathcal{B}(\mathbb{R})$. The elements of $\mathcal{B}(\mathbb{R})$ are called **Borel sets**.

- Every open set is a Borel set and since a σ -algebra is closed with respect to the formation of complements, every closed set is also a Borel set.
- Therefore, since each singleton set is closed, every countable set is a Borel set.

Definition 1.3.6: F_σ and G_δ Sets

A set that is a countable union of closed sets is called an F_σ set, and a set that is a countable intersection of open sets is called a G_δ set.

- Therefore, each F_σ set is a Borel set, and each G_δ set is a Borel set.
- Moreover, both the $\limsup_{n \rightarrow \infty} A_n$ and $\liminf_{n \rightarrow \infty} A_n$ of a sequence of Borel sets are Borel sets.

1.4 Sequence of Real Numbers

We now review the concept of a sequence of real numbers, which is a fundamental concept in analysis.

Definition 1.4.1: Sequence of Real Numbers

A sequence of real numbers is a function $f : \mathbb{N} \rightarrow \mathbb{R}$, which we denote by $\{x_n\}_{n=1}^\infty$, where $x_n = f(n)$.

A sequence $\{a_n\}$ is said to **converge** to a real number a if for every $\epsilon > 0$, there exists a positive integer N such that for all $n \geq N$, we have $|a_n - a| < \epsilon$. In this case, we write $\lim_{n \rightarrow \infty} a_n = a$ or $a_n \rightarrow a$ as $n \rightarrow \infty$.

A **Cauchy sequence** is a sequence $\{a_n\}$ such that for every $\epsilon > 0$, there exists a positive integer N such that for all $m, n \geq N$, we have $|a_n - a_m| < \epsilon$.

- **Monotone Convergence Theorem:** A monotone sequence of real numbers converges if and only if it is bounded.
- **The Bolzano-Weierstrass Theorem:** Every bounded sequence of real numbers has a convergent subsequence.
- **Cauchy Criterion:** A sequence of real numbers converges if and only if it is a Cauchy sequence.
- **Linearity and Montonicity of Convergence of Real Sequences:** Let $\{a_n\}$ and $\{b_n\}$ be two convergent sequences of real numbers, with limits a and b respectively. Then for every real number α, β , the sequence $\{\alpha a_n + \beta b_n\}$ converges to $\alpha a + \beta b$. If $a_n \leq b_n$ for all n , then $a \leq b$.

Convergence to infinity is defined as follows: A sequence $\{a_n\}$ converges to $+\infty$ if for every real number M , there exists a positive integer N such that for all $n \geq N$, we have $a_n > M$. Similarly, $\{a_n\}$ converges to $-\infty$ if for every real number M , there exists a positive integer N such that for all $n \geq N$, we have $a_n < M$.

The **Limit Superior** and **Limit Inferior** of a sequence $\{a_n\}$ are defined as follows:

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} a_k, \quad \liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} a_k. \quad (1.8)$$

Let $\{a_n\}$ and $\{b_n\}$ be two sequences of real numbers. We have

- $\limsup_{n \rightarrow \infty} \{a_n\} = l \in \mathbb{R}$ if and only if for every $\epsilon > 0$, there are infinitely many n such that $a_n > l - \epsilon$ and only finitely many n such that $a_n > l + \epsilon$.
- $\liminf_{n \rightarrow \infty} \{a_n\} = \infty$ if and only if the sequence $\{a_n\}$ is not bounded above.
- The equality:

$$\limsup_{n \rightarrow \infty} \{a_n\} = -\liminf_{n \rightarrow \infty} \{-a_n\} \quad (1.9)$$

- The sequence $\{a_n\}$ converges to an extended real number a if and only if

$$\limsup_{n \rightarrow \infty} \{a_n\} = \liminf_{n \rightarrow \infty} \{a_n\} = a \quad (1.10)$$

- If $a_n \leq b_n$ for all n , then

$$\limsup_{n \rightarrow \infty} \{a_n\} \leq \limsup_{n \rightarrow \infty} \{b_n\}, \quad \liminf_{n \rightarrow \infty} \{a_n\} \leq \liminf_{n \rightarrow \infty} \{b_n\} \quad (1.11)$$

The same logic can be applied to sequences of sets, as we have seen in the previous section.

A **series** is a sequence of partial sums of a sequence $\{a_n\}$, denoted by $\sum_{n=1}^{\infty} a_n$. The series converges to a real number s if the sequence of partial sums converges to s , that is, if

$$\lim_{N \rightarrow \infty} \sum_{n=1}^N a_n = s. \quad (1.12)$$

1.5 Continuous Real-Valued Functions

Let f be a real-valued function defined on a subset E of $\mathbb{R}$. We say that f is continuous at a point $x_0 \in E$ if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x \in E$ with $|x - x_0| < \delta$, we have $|f(x) - f(x_0)| < \epsilon$. If f is continuous at every point in E , we say that f is continuous on E .

We say that f is **Lipchitz continuous** on E if there exists a constant $c > 0$ such that for all $x, y \in E$, we have $|f(x) - f(y)| \leq c|x - y|$. It is clear that a Lipschitz function is continuous, but the converse is not true in general (take $\sqrt{x}$ as an example).

We say that f is **uniformly continuous** on E if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x, y \in E$ with $|x - y| < \delta$, we have $|f(x) - f(y)| < \epsilon$. It is clear that a uniformly continuous function is continuous, but the converse is not true in general (take $\frac{1}{x}$ on $(0, 1)$ as an example).

Proposition:

A real-valued function f defined on $E \subseteq \mathbb{R}$ is continuous at $x_* \in E$ if and only if for every sequence $\{x_n\}$ in E that converges to x_* , the sequence $\{f(x_n)\}$ converges to $f(x_*)$.

f is continuous on E if and only if for every open set $\mathcal{O} \subseteq \mathbb{R}$, the preimage $f^{-1}(\mathcal{O}) = \{x \in E : f(x) \in \mathcal{O}\}$ is open in E .

The Extreme Value Theorem: If f is continuous on a closed and bounded interval $[a, b]$, then f attains its maximum and minimum values on $[a, b]$.

The Intermediate Value Theorem: If f is continuous on a closed interval $[a, b]$ and $f(a) < f(b)$, then for every y between $f(a)$ and $f(b)$, there exists a point $c \in (a, b)$ such that $f(c) = y$.

The Uniform Continuity Theorem: If f is continuous on a closed and bounded interval $[a, b]$, then f is uniformly continuous on $[a, b]$.

Chapter 2

Lebesgue Measure

2.1 Introduction

The Riemann integral of a bounded function over a closed, bounded interval is defined using approximations of the function that are associated with partitions of its domain into finite collections of subintervals. However, this approach has limitations, particularly when dealing with functions that are not continuous or when the domain is not a closed interval. The Lebesgue integral extends the concept of integration to a broader class of functions and domains by focusing on the measure of sets rather than the partitions of intervals.

The generalization of the Riemann integral to the Lebesgue integral will be achieved by using approximations of the function that are associated with decompositions of its domain into finite collections of sets which we call Lebesgue measurable. Each interval is Lebesgue measurable. The richness of the collection of Lebesgue measurable sets provides better upper and lower approximations of a function, and therefore of its integral, than are possible by just employing intervals. This leads to a larger class of functions that are Lebesgue integrable over very general domains and an integral that has better properties.

Remark:

The **length** of an interval is defined as the difference between its endpoints if it is bounded, and as infinity if it is unbounded. Length is an example of a set function, that is, a function that associates an extended real number to each set in a collection of sets. In the case of length, the domain is the collection of all intervals. In this chapter we extend the set function length to a large collection of sets of real numbers.

For instance, the “length” of an open set will be the sum of the lengths of the countable number of open intervals of which it is composed. However, the collection of sets consisting of intervals and open sets is still too limited for our purposes. We will construct a collection of sets that is called **the Lebesgue measurable sets** and a set function called **the Lebesgue measure**, denoted by m .

Definition 2.1.1: Lebesgue Measurable Sets

The collection of **Lebesgue measurable sets** is a σ -algebra of subsets of $\mathbb{R}$ that contains all open sets and closed sets. This conditions are not enough to define the collection of Lebesgue measurable sets, but we will give a more precise definition later in this chapter

by Carathéodory's criterion for measurability.

Definition 2.1.2: Lebesgue Measure

The **Lebesgue measure** is a set function m defined on the collection of Lebesgue measurable sets that satisfies the following properties:

- Each nonempty interval is Lebesgue measurable and its measure is equal to its length.
- Measure is translation invariant: if E is Lebesgue measurable and $x \in \mathbb{R}$, then $E + x = \{y + x : y \in E\}$ is Lebesgue measurable and $m(E + x) = m(E)$.
- Measure is countably additive over disjoint sets: if $\{E_n\}_{n=1}^{\infty}$ is a countable collection of pairwise disjoint Lebesgue measurable sets, then $\bigcup_{n=1}^{\infty} E_n$ is Lebesgue measurable and $m\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} m(E_n)$.

It is not possible to construct a set function that possesses the above three properties and is defined for all sets of real numbers. We respond to this limitation by constructing a set function on a very rich class of sets that does possess the above three properties. The construction has two stages.

We first construct a set function called the **outer measure**, denoted by m^* , which is defined for all sets of real numbers and possesses the first two properties above: outer measure is translation invariant and the outer measure of an interval is equal to its length. However, outer measure is not finitely additive. But it is countably subadditive, which means that if $\{E_n\}_{n=1}^{\infty}$ is a countable collection of sets of real numbers, then

$$m^*\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} m^*(E_n). \quad (2.1)$$

The second stage in the construction is to determine what it means for a set to be Lebesgue measurable. We then restrict the set function m^* to the collection of Lebesgue measurable sets to obtain the Lebesgue measure m .

2.2 Lebesgue Outer Measure

We define the length of an interval I as follows:

$$\ell(I) = \begin{cases} b - a & \text{if } I = [a, b], (a, b), [a, b), (a, b] \\ \infty & \text{if } I = [a, \infty), (a, \infty), (-\infty, b], (-\infty, b), (-\infty, \infty) \end{cases} \quad (2.2)$$

Consider a set $A \subseteq \mathbb{R}$, consider the countable collection of nonempty bounded open intervals $\{I_n\}_{n=1}^{\infty}$ that cover A , that is, $A \subseteq \bigcup_{n=1}^{\infty} I_n$. For each such collection, consider the sum of the lengths of the intervals in the collection. Since the lengths are positive numbers, each sum is uniquely defined independently of the order of the terms. We then take the infimum of all such sums over all countable collections of nonempty bounded open intervals that cover A . This infimum is called the outer measure of A .

Definition 2.2.1: Outer Measure

The **outer measure** of a set $A \subseteq \mathbb{R}$ is defined as

$$m^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \ell(I_n) : A \subseteq \bigcup_{n=1}^{\infty} I_n, I_n \text{ is a nonempty bounded open interval} \right\}. \quad (2.3)$$

It follows immediately from the definition of outer measure that $m^*(\emptyset) = 0$ and is **monotone**, that if $A \subseteq B$, then $m^*(A) \leq m^*(B)$.

Example: **Outer Measure****A countable set has outer measure zero:**

Proof. Let $A = \{x_1, x_2, x_3, \dots\}$ be a countable set. For each $n \in \mathbb{N}$, let I_n be the open interval $(x_n - \epsilon/2^{n+1}, x_n + \epsilon/2^{n+1})$. Then $\{I_n\}_{n=1}^{\infty}$ is a countable collection of nonempty bounded open intervals that cover A . The sum of the lengths of the intervals in this collection is

$$\sum_{n=1}^{\infty} \ell(I_n) = \sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \epsilon. \quad (2.4)$$

Since $\epsilon > 0$ is arbitrary, it follows that $m^*(A) = 0$. □

Proposition: **Outer Measure of Intervals**

The outer measure of an interval is equal to its length.

Proof. The Closed Bounded Interval: Let $[a, b]$ be a closed bounded interval. Let $\epsilon > 0$, we can cover $[a, b]$ with the open interval $(a - \epsilon, b + \epsilon)$, so we have $m^*([a, b]) \leq b - a$.

The other direction is more difficult. Let $\{I_k\}_{k=1}^{\infty}$ be a countable collection of nonempty bounded open intervals that cover $[a, b]$. Since $[a, b]$ is compact, there exists a finite subcollection $\{I_k\}_{k=1}^n$ that covers $[a, b]$. We study the behavior of the finite sum:

Since $a \in [a, b]$, there exists I_{k_1} such that $a \in I_{k_1}$. Let b_1 be the right endpoint of I_{k_1} . If $b_1 \geq b$, then we have covered the entire interval and we are done. If not, then there exists I_{k_2} such that $b_1 \in I_{k_2}$. Let b_2 be the right endpoint of I_{k_2} and continue this process, as it must terminate after a finite number of steps since we have a finite collection of intervals. We obtain a finite collection of intervals $\{I_{k_1}, I_{k_2}, \dots, I_{k_m}\}$ that covers $[a, b]$ and such that the right endpoint of I_{k_m} is greater than or equal to b . The sum of the lengths of these intervals is greater than or equal to $b - a$, since they cover the interval $[a, b]$. Therefore, we have

$$\sum_{k=1}^n \ell(I_k) \geq b - a.$$

So we have $m^*([a, b]) \geq b - a$. Combining this with the previous inequality, we conclude that $m^*([a, b]) = b - a$.

Any Bounded Interval: The proof for any bounded interval follows from the fact that take any bounded interval, for any $\epsilon > 0$ there exists closed bounded interval J_1 and J_2 such that

$J_1 \subseteq I \subseteq J_2$ and $\ell(I) - \epsilon < \ell(J_1)$ and $\ell(J_2) < \ell(I) + \epsilon$. Then we have

$$\ell(I) - \epsilon < m^*(J_1) \leq m^*(I) \leq m^*(J_2) < \ell(I) + \epsilon.$$

which implies that $m^*(I) = \ell(I)$.

Unbounded Interval: The proof for any unbounded interval follows from the fact that take any unbounded interval, for any $M > 0$ there exists a bounded interval J such that $J \subseteq I$ and $\ell(J) > M$. Then we have

$$M < m^*(J) \leq m^*(I).$$

therefore $m^*(I) = \infty$. □

Proposition: **Outer Measure is Translation Invariant**

The outer measure is translation invariant, that is, if $A \subseteq \mathbb{R}$ and $x \in \mathbb{R}$, then $m^*(A + x) = m^*(A)$.

Proof. Simply follows that if $\{I_n\}_{n=1}^\infty$ is a countable collection of nonempty bounded open intervals that cover A , then $\{I_n + x\}_{n=1}^\infty$ is a countable collection of nonempty bounded open intervals that cover $A + x$ and they have the same total length. Therefore, we have $m^*(A + x) = m^*(A)$. □

Proposition: **Outer Measure is Countably Subadditive**

The outer measure is countably subadditive, that is, if $\{E_n\}_{n=1}^\infty$ is a countable collection of sets of real numbers, then

$$m^*\left(\bigcup_{n=1}^\infty E_n\right) \leq \sum_{n=1}^\infty m^*(E_n). \quad (2.5)$$

Proof. If one of E_n has infinite outer measure, then the inequality is trivial. So we assume that $m^*(E_n) < \infty$ for all $n \in \mathbb{N}$.

We begin by noting that from the definition of outer measure, for each $\epsilon > 0$, there exists a countable collection of nonempty bounded open intervals $\{I_{n,k}\}_{k=1}^\infty$ that cover E_n such that their total length $\sum_{k=1}^\infty \ell(I_{n,k}) < m^*(E_n) + \epsilon/2^n$. Then we have the union of all these intervals $\{I_{n,k}\}_{n,k=1}^\infty$ is a countable collection of nonempty bounded open intervals that cover $\bigcup_{n=1}^\infty E_n$. So we have

$$m^*\left(\bigcup_{n=1}^\infty E_n\right) \leq \sum_{n=1}^\infty \sum_{k=1}^\infty \ell(I_{n,k}) < \sum_{n=1}^\infty m^*(E_n) + \epsilon.$$

which completes the proof since $\epsilon > 0$ is arbitrary. □

We can also take $E_n = \emptyset$ for all n sufficiently large, to get the finite version of the above proposition, which is called **finite subadditivity**.

2.3 The σ -algebra of Lebesgue Measurable Sets

Outer measure has four virtues:

- It is defined for all sets of real numbers.

- The outer measure of an interval is equal to its length.
- It is countably subadditive.
- It is translation invariant.

But outer measure fails to be countably additive. In fact, it is not even finitely additive: there are disjoint sets A and B such that $m^*(A \cup B) < m^*(A) + m^*(B)$.

To ameliorate this fundamental defect we identify a σ -algebra of sets, called the Lebesgue measurable sets, which contains all intervals and all open sets and has the property that the restriction of the set function outer measure to the collection of Lebesgue measurable sets is countably additive. There are a number of ways to define what it means for a set to be measurable.

We follow an approach due to Constantine Carathéodory, which is based on the following idea: a set E is Lebesgue measurable if for every set A , the outer measure of A is equal to the sum of the outer measures of the parts of A that are inside and outside of E . In other words

Definition 2.3.1: Carathéodory's Criterion for Measurability

A set $E \subseteq \mathbb{R}$ is **Lebesgue measurable** if for every set $A \subseteq \mathbb{R}$, we have

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c). \quad (2.6)$$

Remark:

To truly understand why the outer measure m^* fails to be additive, we must look at the “economical” nature of its definition. Recall that $m^*(E)$ is an *infimum*—it represents the cheapest possible cost (in total length) to cover a set E using a union of open intervals.

The “Double-Dipping” Phenomenon

The fundamental reason additivity fails ($m^*(A \cup B) < m^*(A) + m^*(B)$) is that open intervals are “blunt” instruments. If two disjoint sets A and B are extremely *intertwined* or *tangled* (like the rational and irrational numbers), they may occupy the same general “region” of space without sharing any specific points.

When we measure them separately:

- To cover A , we must use intervals that inevitably contain a lot of “empty space” around the points of A .
- To cover B , we similarly use intervals that contain empty space around the points of B .

Because A and B are intertwined, the “empty space” we used to cover A likely already contains the points of B . When we calculate $m^*(A \cup B)$, the outer measure is “smart” and uses one set of intervals to cover both simultaneously. In this sense, $m^*(A \cup B)$ is a “discounted price.” However, the sum $m^*(A) + m^*(B)$ forces us to pay for that same overlapping space twice. Thus, the sum of the parts appears larger than the whole.

Disjointness vs. Metric Separation

In elementary geometry, we assume that if two things don’t touch, they are separate. In Real Analysis, this is not true. Two sets can have an intersection that is empty ($A \cap B = \emptyset$) while

their *distance* is zero ($\text{dist}(A, B) = 0$). Outer measure is perfectly additive for sets that have a physical gap between them. The failure of additivity only occurs when sets are “infinitely close” at every point, a condition that occurs with non-measurable sets like the Vitali set.

The “Clean Slicer” Analogy for Carathéodory

Carathéodory’s Criterion ($\forall A : m^*(A) = m^*(A \cap E) + m^*(A \cap E^c)$) defines a measurable set E as a set that acts as a **perfectly sharp knife**.

Most sets in $\mathbb{R}$ are “fuzzy” or “porous” at a microscopic level. If you try to use a “fuzzy” set E to slice a “test set” A , the boundary of E will “fray” the edges of A . This fraying creates the overlap discussed above, causing $m^*(A \cap E) + m^*(A \cap E^c)$ to become larger than the original $m^*(A)$.

A set E is **measurable** if it is “solid” enough that it can slice *any* other set A (no matter how complicated or non-measurable A itself is) without creating any new overlapping covers. We use an arbitrary “test set” A because we want to ensure E is well-behaved everywhere. If E can partition *every* possible set in the universe without “adding” fake volume via overlapping covers, then E is stable enough to belong to our σ -algebra.

Summary of the Logic

Non-additivity is a byproduct of the fact that we cover sets with **open intervals**. Because open intervals have “insides,” they cannot perfectly conform to the jagged, “point-cloud” nature of non-measurable sets. Carathéodory’s definition filters out these jagged sets, leaving us with a collection (the Lebesgue sets) where the “bluntness” of our intervals no longer causes mathematical errors.

It is easy to see that this means that if E is Lebesgue measurable then for every other set A that is disjoint from E , we have

$$m^*(A \sqcup E) = m^*(A) + m^*(E),$$

just taking the test set to be $A \sqcup E$ would do.

Observe that the definition of measurability is symmetric in E and E^c . Therefore, if E is Lebesgue measurable, then E^c is also Lebesgue measurable. And clearly, $\mathbb{R}$ and $\emptyset$ are Lebesgue measurable.

Proposition:

Any set of outer measure zero is measurable. In particular, any countable set is measurable.

Proof. Let E be a set of outer measure zero. Then for any set A , we have

$$m^*(A) \leq m^*(A \cap E) + m^*(A \cap E^c) \leq m^*(E) + m^*(A) = 0 + m^*(A) = m^*(A).$$

Therefore, E is measurable. □

Proposition:

The union of a finite collection of measurable sets is measurable.

Proof. Just give the case for two sets, the general case follows by induction. Let E_1 and E_2 be two measurable sets. Then for any set A , we try to partition A into four parts: $A \cap E_1 \cap E_2$, $A \cap E_1 \cap E_2^c$, $A \cap E_1^c \cap E_2$, and $A \cap E_1^c \cap E_2^c$. Then we first use the measurability of E_1 to get

$$m^*(A) = m^*(A \cap E_1) + m^*(A \cap E_1^c).$$

and then we use the measurability of E_2 and let the test set be $A \cap E_1$ and $A \cap E_1^c$ respectively to get

$$\begin{aligned} m^*(A \cap E_1) &= m^*(A \cap E_1 \cap E_2) + m^*(A \cap E_1 \cap E_2^c), \\ m^*(A \cap E_1^c) &= m^*(A \cap E_1^c \cap E_2) + m^*(A \cap E_1^c \cap E_2^c). \end{aligned}$$

Adding these two equations together, and combine the three terms on the right hand side and use subadditivity to get

$$m^*(A) \geq m^*(A \cap (E_1 \cup E_2)) + m^*(A \cap (E_1 \cup E_2)^c).$$

The equality in the other direction follows from direct application of subadditivity. Therefore, $E_1 \cup E_2$ is measurable. $\square$

Proposition: Additivity of Outer Measure on Measurable Sets

If $\{E_k\}_{k=1}^n$ is a finite collection of pairwise disjoint measurable sets, then for any set A , we have

$$m^*\left(A \cap \bigcup_{k=1}^n E_k\right) = \sum_{k=1}^n m^*(A \cap E_k). \quad (2.7)$$

In particular, if $A = \bigcup_{k=1}^n E_k$, then we have

$$m^*\left(\bigcup_{k=1}^n E_k\right) = \sum_{k=1}^n m^*(E_k). \quad (2.8)$$

Proof. Again, we prove the case for $n = 2$, the general case follows by induction. Let E_1 and E_2 be two disjoint measurable sets. Then for any set A , we have

$$m^*(A) = m^*(A \cap (E_1 \cup E_2)) + m^*(A \cap (E_1 \cup E_2)^c).$$

and also according to above proposition, we have

$$m^*(A) = m^*(A \cap (E_1 \cap E_2^c)) + m^*(A \cap (E_1^c \cap E_2)) + m^*(A \cap (E_1^c \cap E_2^c)).$$

as they are disjoint, using the outer measure of empty set is zero. We can simplify the above equation to get

$$m^*(A) = m^*(A \cap E_1) + m^*(A \cap E_2) + m^*(A \cap (E_1 \cup E_2)^c).$$

So we have

$$m^*(A \cap (E_1 \cup E_2)) = m^*(A \cap E_1) + m^*(A \cap E_2).$$

as desired. The second part of the proposition follows by taking $A = E_1 \cup E_2$. $\square$

Definition 2.3.2: Algebra

A collection $\mathcal{A}$ of subsets of a set X is called an **algebra** if it satisfies the following properties:

- $X \in \mathcal{A}$.
- If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$.
- If $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$.

Remark:

The difference of an algebra and a σ -algebra is that an algebra is closed under finite unions, while a σ -algebra is closed under countable unions. Therefore, every σ -algebra is an algebra, but not vice versa.

We have thereby established that the collection of Lebesgue measurable sets is an algebra, and that the restriction of the outer measure to this algebra is finitely additive.

To proceed the discussion, we need to adopt a useful trick to turn a countable union of sets into a disjoint union. Consider $\{E_n\}_{n=1}^\infty$ be a countable collection of sets. We can define a new collection of sets $\{E'_n\}_{n=1}^\infty$ by

$$E'_1 = E_1, \quad E'_n = E_n \setminus \bigcup_{k=1}^{n-1} E_k, \quad n \geq 2. \quad (2.9)$$

As Lebesgue measurable sets form an algebra, we have that E'_n is Lebesgue measurable for all $n \in \mathbb{N}$. It is easy to see that $\{E'_n\}_{n=1}^\infty$ is a countable collection of pairwise disjoint sets and have the same partial union as $\{E_n\}_{n=1}^\infty$, that is, $\bigcup_{k=1}^n E'_k = \bigcup_{k=1}^n E_k$ for all $n \in \mathbb{N}$.

Proposition: **Countable Union of Measurable Sets is Measurable**

The union of a countable collection of measurable sets is measurable.

Proof. As said above, we may just assume that the collection of sets $\{E_n\}_{n=1}^\infty$ is pairwise disjoint. Take $F_n = \bigcup_{k=1}^n E_k$, and $E = \bigcup_{k=1}^\infty E_k$. Then F_n is measurable for all $n \in \mathbb{N}$, and we need to show that E is measurable. For any set A , we have

$$m^*(A) = m^*(A \cap F_n) + m^*(A \cap F_n^c) \geq m^*(A \cap F_n) + m^*(A \cap E^c). \quad (2.10)$$

Consider the sequence $\{m^*(A \cap F_n)\}_{n=1}^\infty$. It is a nondecreasing sequence with an upper bound $m^*(A \cap E)$. Therefore, we have

$$m^*(A) \geq \lim_{n \rightarrow \infty} m^*(A \cap F_n) + m^*(A \cap E^c) = m^*(A \cap E) + m^*(A \cap E^c). \quad (2.11)$$

and the other direction of the inequality follows from subadditivity. Therefore, E is measurable. $\square$

We have now established that the collection of Lebesgue measurable sets is a σ -algebra.

Proposition: **Interval is Measurable**

Every interval is measurable.

Proof. We need only prove that (a, ∞) is measurable, as the other cases follow from this one by intersection and complementation.

For any set A , assume $a \notin A$, otherwise we can just remove a from A without changing the outer measure. Partition A into two parts: $A_1 = A \cap (-\infty, a)$ and $A_2 = A \cap (a, \infty)$. Then consider any open bounded intervals $\{I_k\}_{k=1}^{\infty}$ that cover A , we partition them into two parts: $I'_k = I_k \cap (-\infty, a)$ and $I''_k = I_k \cap (a, \infty)$, each of which is an open bounded interval that covers A_1 and A_2 respectively. Then we have

$$\ell(I_k) = \ell(I'_k) + \ell(I''_k).$$

and by definition,

$$m^*(A_1) \leq \sum_{k=1}^{\infty} \ell(I'_k), \quad m^*(A_2) \leq \sum_{k=1}^{\infty} \ell(I''_k).$$

which gives

$$m^*(A_1) + m^*(A_2) \leq \sum_{k=1}^{\infty} \ell(I_k).$$

And as $m^*(A)$ is the infimum of all such sums, we have

$$m^*(A_1) + m^*(A_2) \leq m^*(A). \quad (2.12)$$

so we have our desired result. $\square$

As every open set is a countable union of open intervals, we have that every open set is measurable. And as the complement of an open set is closed, we have that every closed set is measurable.

We have now established that the collection of Lebesgue measurable sets is a σ -algebra that contains all open sets and closed sets. This means that it contains the Borel σ -algebra, which is the smallest σ -algebra that contains all open and closed sets. Therefore, every Borel set is Lebesgue measurable.

Proposition: **Translation Invariance of Measurable Sets**

If E is a measurable set and $x \in \mathbb{R}$, then $E + x = \{y + x : y \in E\}$ is measurable.

Proof. Easy, just move the test set A by $-x$ and use the translation invariance of outer measure:

$$m^*(A) = m^*(A - x) = m^*((A - x) \cap E) + m^*((A - x) \cap E^c) = m^*(A \cap (E + x)) + m^*(A \cap (E + x)^c).$$

which shows that $E + x$ is measurable. $\square$

2.4 Outer and Inner Approximations of Measurable Sets

We now present two characterizations of measurability of a set, one based on inner approximation by closed sets and the other on outer approximation by open sets, which provide alternate angles

of vision on measurability. These characterizations will be essential tools for our forthcoming study of approximation properties of measurable and integrable functions.

Measurable sets have a **Excision Property**, which means that if A is a measurable set with finite outer measure, and $A \subseteq B$, then we have

$$m^*(B \setminus A) = m^*(B) - m^*(A). \quad (2.13)$$

which is quite useful but very obvious from the definition. The finiteness of the outer measure of A is necessary for the subtraction to make sense.

Theorem 2.4.1: Equivalent Characterizations of Measurability

Let $E \subseteq \mathbb{R}$ be a set. The following statements are equivalent to E being Lebesgue measurable:

1. E is Lebesgue measurable.
2. For every $\epsilon > 0$, there exists an open set $\mathcal{O}$ such that $E \subseteq \mathcal{O}$ and $m^*(\mathcal{O} \setminus E) < \epsilon$.
3. There exists a G_δ set G such that $E \subseteq G$ and $m^*(G \setminus E) = 0$.
4. For every $\epsilon > 0$, there exists a closed set F such that $F \subseteq E$ and $m^*(E \setminus F) < \epsilon$.
5. There exists an F_σ set F such that $F \subseteq E$ and $m^*(E \setminus F) = 0$.

Proof. Proving the first two statements would suffice, as the other two statements follow from the first two by taking complements.

(1) $\Rightarrow$ (2) Assume E is measurable. Let $\epsilon > 0$. First consider the case that $m^*(E) < \infty$. Then by the definition we can find a countable collection of nonempty bounded open intervals $\{I_n\}_{n=1}^\infty$ that cover E such that $\sum_{n=1}^\infty \ell(I_n) < m^*(E) + \epsilon$. Let $\mathcal{O} = \bigcup_{n=1}^\infty I_n$, then we have $E \subseteq \mathcal{O}$ and $m^*(\mathcal{O}) \leq \sum_{n=1}^\infty \ell(I_n) < m^*(E) + \epsilon$, which is the desired result.

Now consider the case that $m^*(E) = \infty$. Then we express E as a countable disjoint union of measurable sets with finite outer measure. This can be accomplished by taking $E_n = E \cap [-n, n]$ for $n \in \mathbb{N}$ and making them disjoint by subtracting the previous sets. Then for the finite case, we let

$$m^*(\mathcal{O}_k \setminus E_k) < \epsilon/2^k, \quad k = 1, 2, \dots$$

Then we have $\mathcal{O} = \bigcup_{k=1}^\infty \mathcal{O}_k$ is an open set that contains E and $\mathcal{O} \setminus E = \bigcup_{k=1}^\infty (\mathcal{O}_k \setminus E_k)$. Then by subadditivity, we have

$$m^*(\mathcal{O} \setminus E) \leq \sum_{k=1}^\infty m^*(\mathcal{O}_k \setminus E_k) < \sum_{k=1}^\infty \epsilon/2^k = \epsilon.$$

(2) $\Rightarrow$ (3) Take a sequence of $\mathcal{O}_k$ such that $m^*(\mathcal{O}_k \setminus E) < 1/k$ for $k = 1, 2, \dots$. Then define $G = \bigcap_{k=1}^\infty \mathcal{O}_k$, which is a G_δ set. Then we have $E \subseteq G$ and

$$m^*(G \setminus E) \leq m^*(\mathcal{O}_k \setminus E) < 1/k, \quad k = 1, 2, \dots$$

implying that $m^*(G \setminus E) = 0$.

(3) $\Rightarrow$ (1) Let G be a G_δ set such that $E \subseteq G$ and $m^*(G \setminus E) = 0$. Then both a G_δ set and a set of outer measure zero are measurable, so E is measurable. $\square$

Remark:

This theorem shows that a set is measurable if and only if an outer approximation by open sets can be chosen so that it does not cover too much extra space, or equivalently, if and only if an inner approximation by closed sets can be chosen so that it does not miss too much of the set. This is a very useful characterization of measurability.

The following property of measurable sets of finite outer measure asserts that such sets are nearly equal to the disjoint union of a finite number of open intervals.

Theorem 2.4.2: Littlewood's First Principle

Let $E \subseteq \mathbb{R}$ be a measurable set with finite outer measure. Then for every $\epsilon > 0$, there exists a finite collection of disjoint open intervals $\{I_k\}_{k=1}^n$ such that if $\mathcal{O} = \bigcup_{k=1}^n I_k$, then

$$m^*(\mathcal{O} \setminus E) + m^*(E \setminus \mathcal{O}) < \epsilon. \quad (2.14)$$

Proof. This is quite straightforward from the previous theorem. Let $\epsilon > 0$. Then there exists an open set $\mathcal{U}$ such that $E \subseteq \mathcal{U}$ and $m^*(\mathcal{U} \setminus E) < \epsilon/2$. Then we can express $\mathcal{U}$ as a countable union of disjoint open intervals $\{I_k\}_{k=1}^\infty$. Since we have

$$\sum_{k=1}^n \ell(I_k) = m^*\left(\bigcup_{k=1}^n I_k\right) \leq m^*(\mathcal{U}) < \infty$$

we can find n sufficiently large such that

$$\sum_{k=n+1}^\infty \ell(I_k) < \epsilon/2.$$

the remaining intervals $\{I_k\}_{k=1}^n$ give the desired result. $\square$

2.5 Countable Additivity, Continuity, and Borel-Cantelli Lemma

We hereby fill the last puzzle piece of the theory of Lebesgue measure by proving that the restriction of outer measure to the σ -algebra of Lebesgue measurable sets is countably additive.

Definition 2.5.1: Lebesgue Measure

The restriction of outer measure to the σ -algebra of Lebesgue measurable sets is called **Lebesgue measure**, and is denoted by m .

Proposition: **Countable Additivity of Lebesgue Measure**

If $\{E_n\}_{n=1}^\infty$ is a countable collection of pairwise disjoint measurable sets, then

$$m\left(\bigcup_{n=1}^\infty E_n\right) = \sum_{n=1}^\infty m(E_n). \quad (2.15)$$

Proof. We have already proved that the union of a countable collection of measurable sets is measurable, so the left hand side is well-defined. The finite version and the monotonicity of outer measure shows that

$$m\left(\bigcup_{k=1}^\infty E_k\right) \geq m\left(\bigcup_{k=1}^n E_k\right) = \sum_{k=1}^n m(E_k), \quad n = 1, 2, \dots$$

so we have

$$m\left(\bigcup_{k=1}^\infty E_k\right) \geq \sum_{k=1}^\infty m(E_k).$$

the equality follows from the countable subadditivity of outer measure. $\square$

Theorem 2.5.1: The Lebesgue Measure

The set function Lebesgue measure, defined on the σ -algebra of Lebesgue measurable sets, assigns length to any interval, is translation invariant, and is countable additive.

Remark:

It is natural to ask if the Lebesgue measurable sets $\mathcal{M}$ constructed above is the largest possible σ -algebra that admits a Lebesgue measure. The answer depends on our requirements:

- **Existence:** By the Vitali Theorem, no such measure can exist on the full power set $\mathcal{P}(\mathbb{R})$.
- **Canonicity:** The Lebesgue σ -algebra $\mathcal{M}$ is the **largest** collection of sets that can be derived canonically from the outer measure m^* using the Carathéodory criterion.
- **Extensions:** While it is possible to extend the measure to even larger σ -algebras using the Axiom of Choice, such extensions are not unique and lack the structural "elegance" required for standard analysis.

Thus, for all practical purposes in analysis and probability, $\mathcal{M}$ is treated as the maximal domain for the Lebesgue measure.

Some would say the existence of larger σ -algebras with a Lebesgue measure is not quite surprising when we look at the Carathéodory criterion. The criterion itself gives a stronger requirements that a measurable set must give a good partition to ANY other set, which is a very strong requirement, compared to the natural requirement that a measurable set need only give a good partition to other measurable sets. This is not quite right, as one can prove that we can weaken the Carathéodory criterion to only require that a measurable set gives a good partition to other OPEN sets, and this is equivalent to the original definition.

We may well regard the above construction of the Lebesgue measure as the definition for both Lebesgue measure and Lebesgue measurable sets, and the above theorem as the fun-

damental properties of the Lebesgue measure. But it may feel unsatisfactory that we have not given an axiomatic definition. It is natural to think that the Lebesgue measurable sets are the “maximal” σ -algebra that satisfy some natural properties. As we see that the above three properties are natural but not enough, we need to compel ourself to another restriction.

Definition 2.5.2: Lebesgue Measurable Sets and Lebesgue Measure

The **Lebesgue measurable sets** $\mathcal{M}$ and the **Lebesgue measure** m are defined as the unique maximal σ -algebra and its associated set function $m : \mathcal{M} \rightarrow [0, \infty]$ such that:

- m is a **measure** (it is countably additive over disjoint sets).
- $\mathcal{M}$ contains all intervals and $m(I) = \ell(I)$.
- m is **translation invariant**.
- m is **consistent with outer measure**: $m(E) = m^*(E)$ for all $E \in \mathcal{M}$.

Next we move on to discuss the properties of the Lebesgue measure.

Theorem 2.5.2: Continuity of Measure

Lebesgue measure possesses the following continuity properties:

- If $\{A_k\}_{k=1}^{\infty}$ is an ascending sequence of measurable sets, then

$$m\left(\bigcup_{k=1}^{\infty} A_k\right) = \lim_{k \rightarrow \infty} m(A_k). \quad (2.16)$$

- If $\{A_k\}_{k=1}^{\infty}$ is a descending sequence of measurable sets with $m(A_1) < \infty$, then

$$m\left(\bigcap_{k=1}^{\infty} A_k\right) = \lim_{k \rightarrow \infty} m(A_k). \quad (2.17)$$

Proof. First we prove the first statement. If there is one k such that $m(A_k) = \infty$, then the result is trivial. Otherwise, we have $m(A_k) < \infty$ for all k . Let $C_k = A_k \setminus A_{k-1}$ for $k \geq 2$ and $C_1 = A_1$. Then $\{C_k\}_{k=1}^{\infty}$ is a countable collection of pairwise disjoint measurable sets, and we have

$$\bigcup_{k=1}^{\infty} A_k = \bigcup_{k=1}^{\infty} C_k.$$

so we have

$$m\left(\bigcup_{k=1}^{\infty} A_k\right) = m\left(\bigcup_{k=1}^{\infty} C_k\right) = \sum_{k=1}^{\infty} m(C_k) = \lim_{n \rightarrow \infty} \sum_{k=1}^n m(C_k) = \lim_{n \rightarrow \infty} m(A_n).$$

The second part of the theorem follows from the first part by taking complements. $\square$

Definition 2.5.3: Almost Everywhere

A property P is said to hold **almost everywhere** (a.e.) on a set E if the set of points in E where P fails has Lebesgue measure zero. In other words, if $F = \{x \in E : P(x) \text{ does not hold}\}$, then $m(F) = 0$.

Lemma 2.5.1: The Borel-Cantelli Lemma

Let $\{E_n\}_{n=1}^{\infty}$ be a sequence of measurable sets. If $\sum_{n=1}^{\infty} m(E_n) < \infty$, then the set of points that belong to infinitely many E_n has measure zero. In other words, if we define

$$E = \limsup_{n \rightarrow \infty} E_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k, \quad (2.18)$$

then $m(E) = 0$.

Proof. As we have

$$m\left(\bigcup_{k=n}^{\infty} E_k\right) \leq \sum_{k=n}^{\infty} m(E_k) < \infty$$

Hence, by the continuity of measure,

$$m(E) = \lim_{n \rightarrow \infty} m\left(\bigcup_{k=n}^{\infty} E_k\right) \leq \lim_{n \rightarrow \infty} \sum_{k=n}^{\infty} m(E_k) = 0.$$

so we have $m(E) = 0$. □

The set function Lebesgue measure inherits the properties possessed by Lebesgue outer measure. For future reference we name some of these properties.

- **Finite Additivity:** If $E_1, E_2, \dots, E_n$ are pairwise disjoint measurable sets, then

$$m\left(\bigcup_{k=1}^n E_k\right) = \sum_{k=1}^n m(E_k). \quad (2.19)$$

- **Monotonicity:** If $E_1 \subseteq E_2$ are measurable sets, then

$$m(E_1) \leq m(E_2). \quad (2.20)$$

- **Excision Property:** If $E_1 \subseteq E_2$ are measurable sets with $m(E_1) < \infty$, then

$$m(E_2 \setminus E_1) = m(E_2) - m(E_1). \quad (2.21)$$

- **Countable Monotonicity:** If $\{E_n\}_{n=1}^{\infty}$ is a countable collection of measurable sets, then

$$m\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} m(E_n). \quad (2.22)$$

Remark:

In our forthcoming study of Lebesgue integration it will be apparent that it is the countable additivity of Lebesgue measure that provides the Lebesgue integral with its decisive advantage over the Riemann integral.

2.6 Non-measurable Sets

We have defined what it means for a set to be measurable and studied properties of the collection of measurable sets. It is only natural to ask if, in fact, there are any sets that fail to be measurable.

It is not quite obvious that measurability does not preserve under subsets. For a set E that has zero measure, any subset of E has outer measure zero and is therefore measurable. But we shall see that this fails for any set that has positive measure.

Lemma 2.6.1: Infinite Packing Lemma

Let E be a bounded measurable set. Suppose there is a bounded, countably infinite set of real numbers Λ such that the translates $\{E + \lambda : \lambda \in \Lambda\}$ are pairwise disjoint. Then $m(E) = 0$.

Proof. This actually means that a finite area cannot be packed by infinite many disjoint copies of itself unless it has zero area, which is quite obvious. We just write

$$m\left(\bigcup_{\lambda \in \Lambda} (E + \lambda)\right) = \sum_{\lambda \in \Lambda} m(E + \lambda) = \sum_{\lambda \in \Lambda} m(E) = \infty$$

unless $m(E) = 0$. □

We define two reals to be **rationally equivalent** if their difference is a rational number. This is an equivalence relation on $\mathbb{R}$, and for any set $E \subseteq \mathbb{R}$, we partition E into equivalence classes under this relation. By the Axiom of Choice, we can select one representative from each equivalence class to form a set $\mathcal{C}_E$, so that

- The difference of any two elements of $\mathcal{C}_E$ is irrational.
- For any $x \in E$, there exists a unique $c \in \mathcal{C}_E$ such that $x - c$ is rational.

Theorem 2.6.1: (Vitali)

Any set E of positive outer measure contains a non-measurable subset.

Proof. Suppose E is bounded, otherwise, from continuity, we can intersect E with a sufficiently large bounded interval to get a bounded set of positive outer measure. Let $\mathcal{C}_E$ be any choice of representatives of the rational equivalence classes of E . Then if it is measurable, let Λ_0 be any bounded, countably infinite set of rational numbers, then the translates $\{\mathcal{C}_E + \lambda : \lambda \in \Lambda_0\}$ are pairwise disjoint and contained in a bounded interval, which means that $m(\mathcal{C}_E) = 0$ by the infinite packing lemma.

Now, using the countable additivity of Lebesgue measure, we have

$$m\left(\bigcup_{\lambda \in \Lambda_0} (\mathcal{C}_E + \lambda)\right) = \sum_{\lambda \in \Lambda_0} m(\mathcal{C}_E + \lambda) = \sum_{\lambda \in \Lambda_0} m(\mathcal{C}_E) = 0.$$

To obtain a contradiction we make a special choice of Λ_0 that make E contained in the union of the translates. Let $E \subseteq [-b, b]$ for some $b > 0$. Let $\Lambda_0 = [-2b, 2b] \cap \mathbb{Q}$, which is a bounded, countably infinite set of rational numbers. This obviously makes $E \subseteq \bigcup_{\lambda \in \Lambda_0} (\mathcal{C}_E + \lambda)$, which implies that

$$m(E) \leq m\left(\bigcup_{\lambda \in \Lambda_0} (\mathcal{C}_E + \lambda)\right) = 0,$$

a contradiction to the assumption. □

Remark:

A direct intuition of the above theorem is that we cannot pack a finite area with infinitely many disjoint copies of something that has a well-defined area.

Proposition:

There are disjoint sets of real numbers A, B such that

$$m^*(A \cup B) < m^*(A) + m^*(B). \quad (2.23)$$

Proof. Otherwise every set follows the Carathéodory criterion and is therefore measurable, which contradicts the existence of non-measurable sets. □

2.7 The Cantor Set and Cantor-Lebesgue Function

We hereby give examples of the relation of countability and zero measure, and the relation between Borel sets and Lebesgue measurable sets. We have already seen, and will see more:

- Every countable set has measure zero, but there are uncountable sets that have measure zero.
- Every Borel set is Lebesgue measurable, but there are Lebesgue measurable sets that are not Borel sets.

The Cantor Set The Cantor set is a classic example of a set that is uncountable yet has measure zero. Visually, it is a fractal-like set that is constructed by repeatedly removing the middle third of intervals:

We start at the closed interval $[0, 1]$. In the first step, we remove the open middle third $(\frac{1}{3}, \frac{2}{3})$, leaving two closed intervals: $[0, \frac{1}{3}]$ and $[\frac{2}{3}, 1]$. In the second step, we remove the middle third of each of these intervals, leaving four intervals: $[0, \frac{1}{9}]$, $[\frac{2}{9}, \frac{1}{3}]$, $[\frac{2}{3}, \frac{7}{9}]$, and $[\frac{8}{9}, 1]$. We continue this process indefinitely. Formally speaking, denote each step as C_k , where $C_0 = [0, 1]$ and C_{k+1} is obtained from C_k by removing the open middle third of each interval in C_k . The Cantor set C is defined as the intersection of all these sets:

$$C = \bigcap_{k=0}^{\infty} C_k. \quad (2.24)$$

Some direct observations about the Cantor set:

- The sequence $\{C_k\}_{k=1}^{\infty}$ is a descending sequence of closed sets.
- For each k , C_k consists of 2^k closed intervals, each of length $1/3^k$.

Theorem 2.7.1: Measurability of the Cantor Set

The Cantor set C is a closed, uncountable set of measure zero.

Proof. The Cantor set is closed as it is the intersection of closed sets. As C_k is closed, so C is closed.

The measure of C_k is $m(C_k) = 2^k/3^k = (2/3)^k$, which tends to zero as $k \rightarrow \infty$. By the continuity of measure, we have $m(C) = \lim_{k \rightarrow \infty} m(C_k) = 0$.

Suppose C is countable. Then we can enumerate its elements as $\{c_k\}_{k=1}^{\infty}$. One of the two intervals in C_1 does not contain c_1 , denote it as F_1 . Then one of the two intervals in F_1 does not contain c_2 , denote it as F_2 . Continuing in this way, we construct a countable collection of sets $\{F_k\}_{k=1}^{\infty}$ such that F_k is a closed interval in C_k that does not contain $c_1, c_2, \dots, c_k$. Then this is a nesting sequence of closed intervals, and by the nested interval theorem, we have $\bigcap_{k=1}^{\infty} F_k \neq \emptyset$. But this intersection does not contain any c_k but is contained in C , which is a contradiction. Therefore, C is uncountable. $\square$

We denote a function to be **increasing** if $x_1 < x_2$ implies $f(x_1) \leq f(x_2)$, and **strictly increasing** if $x_1 < x_2$ implies $f(x_1) < f(x_2)$.

The Cantor-Lebesgue Function We now define the Cantor-Lebesgue function, a continuous, increasing function φ on $[0, 1]$ that despite $\varphi(0) < \varphi(1)$, has derivative zero almost everywhere.

We draft the construction of the Cantor-Lebesgue function φ in a stepwise manner. First, let the middle-thirds part of the interval $[0, 1]$ be mapped to the constant value $1/2$. Then, for the two remaining intervals $[0, 1/3]$ and $[2/3, 1]$, we map each middle-thirds part to the constant values $1/4$ and $3/4$, respectively. In the k -step, we add the definition to the deleted middle-thirds parts of the Cantor set C_k by mapping them to the constant values that is the average of the values of the two side intervals. In this way, we have defined all the values of φ on $\mathcal{O} = [0, 1] \setminus C$.

For the values of φ on the Cantor set C , we extend the definition of φ to C by

$$\varphi(0) = 0, \quad \varphi(x) = \sup\{\varphi(t) : t \in \mathcal{O} \cap [0, x]\}, \quad x \in C. \quad (2.25)$$

Proposition: The Cantor-Lebesgue Function

The Cantor-Lebesgue function φ is an increasing continuous function that maps $[0, 1]$ onto $[0, 1]$. Its derivative exists on the open set $\mathcal{O} = [0, 1] \setminus C$ and is zero on $\mathcal{O}$, which has measure 1.

Proof. Since φ is increasing on $\mathcal{O}$, so is its extension to C .

As for continuity, φ is certainly continuous at $\mathcal{O}$, so we only need to check continuity at C . Let $x \in C$ and $\epsilon > 0$. Then there exists k sufficiently large such that $1/2^k < \epsilon$. Then the nearby staircases of φ are all within $1/2^k$ of $\varphi(x)$, which shows that φ is continuous at x , which shows that φ is continuous on $[0, 1]$.

We use intermediate value theorem to show that φ maps $[0, 1]$ onto $[0, 1]$, just from the continuity of φ and the fact that $\varphi(0) = 0$ and $\varphi(1) = 1$. $\square$

Proposition:

Let φ be the Cantor-Lebesgue function. Then define

$$\psi(x) = \varphi(x) + x, \quad x \in [0, 1]. \quad (2.26)$$

being a strictly increasing continuous function that maps $[0, 1]$ onto $[0, 2]$. It

- maps the Cantor set C onto a set of positive measure, and
- maps a subset of C (a measurable set) onto a non-measurable set.

Proof. For a strictly increasing continuous function, we have $[0, 1] = C \sqcup \mathcal{O}$, and $\psi([0, 1]) = [0, 2] = \psi(C) \sqcup \psi(\mathcal{O})$. We calculate $m(\psi(\mathcal{O}))$:

As $\mathcal{O}$ is a countable union of some $\mathcal{O}_k = [0, 1] \setminus C_k$, we have $\psi(\mathcal{O}_k) = 1 - (2/3)^k$, which means that $m(\psi(\mathcal{O})) = \lim_{k \rightarrow \infty} m(\psi(\mathcal{O}_k)) = 1$. Hence, we have $m(\psi(C)) = m([0, 2]) - m(\psi(\mathcal{O})) = 2 - 1 = 1$, which shows that $\psi(C)$ has positive measure.

For the second part, we can take a non-measurable set $E \subseteq \psi(C)$ from Vitali's theorem, and then $\psi^{-1}(E)$ is a measurable set that is mapped onto a non-measurable set E . $\square$

As ψ is a homeomorphism, it should map Borel sets to Borel sets, as they only concerns topological properties.

Remark:

Well, a Borel set is a topological property, but a Lebesgue measurable set is not. We can see that the definition of Lebesgue measurable sets needs other structure, like a metric system, to describe lengths of intervals, the outer measure and the Carathéodory criterion. Hence, a homeomorphism does not necessarily map Lebesgue measurable sets to Lebesgue measurable sets.

We may not directly see this from the definition as we always concern $\mathbb{R}$ as an embedded example, which has profound structures. We will encounter more abstract discussions in the future, and we will see then that the required structures to establish these theories.

Proposition:

There is a measurable set, a subset of the Cantor set, that is not a Borel set.

Chapter 3

Lebesgue Measurable Functions

We devote this chapter to the study of measurable functions in order to lay the foundation for the study of the Lebesgue integral.

All the functions considered in this chapter take values (codomain) in the extended real numbers, that is, the set $\mathbb{R} \cup \{-\infty, +\infty\}$. This would make the discussion of limits and convergence of functions more convenient. The domain of the functions is assumed to be subsets of $\mathbb{R}$ as usual.

3.1 Sums, Products, and Compositions

Given two functions h, g defined on a measurable set E , we often abbreviate $h < g$ to mean that $h(x) < g(x)$ for all $x \in E$, a sequence $\{f_n\}$ of functions is said to increase provided that $f_n < f_{n+1}$ for all $n \in \mathbb{N}$.

Proposition:

Let the function f have a measurable domain E . Then the following statements are equivalent:

- For every $c \in \mathbb{R}$, the set $\{x \in E : f(x) > c\}$ is measurable.
- For every $c \in \mathbb{R}$, the set $\{x \in E : f(x) < c\}$ is measurable.
- For every $c \in \mathbb{R}$, the set $\{x \in E : f(x) \geq c\}$ is measurable.
- For every $c \in \mathbb{R}$, the set $\{x \in E : f(x) \leq c\}$ is measurable.

and each of these statements implies that the set $\{x \in E : f(x) = c\}$ is measurable for every $c \in \mathbb{R}$.

Proof. We will show that the first statement implies the third, and the rest follows from taking complements. Just noting

$$\{x \in E : f(x) \geq c\} = \bigcap_{n=1}^{\infty} \left\{x \in E : f(x) > c - \frac{1}{n}\right\}$$

would suffice.

Next we show that the first statement implies $\{x \in E : f(x) = c\}$ is measurable. If $c \in \mathbb{R}$ we have

$$\{x \in E : f(x) = c\} = \{x \in E : f(x) \geq c\} \cap \{x \in E : f(x) \leq c\}$$

so it is measurable. If $c = +\infty$ we have

$$\{x \in E : f(x) = +\infty\} = \bigcap_{n=1}^{\infty} \{x \in E : f(x) > n\}$$

so it is measurable. If $c = -\infty$ we follow similar reasoning. $\square$

Definition 3.1.1: Lebesgue Measurable Functions

An extended real-valued function f defined on a measurable set E is said to be **Lebesgue measurable** if it satisfies any of the equivalent conditions in the previous proposition.

Example: Characterization Function

Let E be a measurable set and $A \subseteq E$. The characteristic function χ_A of A is defined by

$$\chi_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

Then χ_A is measurable if and only if A is measurable. This is because $\{x \in E : \chi_A(x) > c\}$ is either A , E , or $\emptyset$, depending on the value of c .

As opens sets in the extended real numbers can be countably generated by intervals, this naturally gives the following equivalent definition:

Proposition: Preimage of Open Sets is Measurable

A function f defined on a measurable set E is Lebesgue measurable if and only if the preimage of every open set in the extended real numbers is measurable.

Remark:

In the context of extended real-valued functions, the topology of $\overline{\mathbb{R}}$ is generated by the open intervals (a, b) , $[-\infty, b)$, and $(a, \infty]$. Thus, an open set in $\overline{\mathbb{R}}$ may contain the points $\pm\infty$. This ensures that the preimage of a neighborhood of infinity, such as $\{x : f(x) > c\} = f^{-1}((c, \infty])$, is treated consistently within the definition of measurability.

The topology of the extended real numbers $\overline{\mathbb{R}}$ contains the standard topology of $\mathbb{R}$. A set $U \subseteq \mathbb{R}$ is open in the standard sense if and only if it is open when viewed as a subset of $\overline{\mathbb{R}}$. This ensures that any Lebesgue measurable function $f : E \rightarrow \overline{\mathbb{R}}$ also behaves correctly when restricted to the set of points where it takes finite values.

The following proposition tells us that the most familiar functions from elementary analysis, the continuous functions, are measurable.

Proposition: Continuous Functions are Measurable

A real-valued function that is continuous on its measurable domain is measurable.

Proof. This is because the preimage of an open set under a continuous function is open, and hence measurable. $\square$

Proposition: Monotone Functions are Measurable

A real-valued function that is monotone on its measurable domain is measurable.

Proof. This is because $f^{-1}((c, \infty))$ is an interval, and hence measurable. $\square$

Proposition:

Let f be an extended real-valued function defined on a measurable set E .

- If f is measurable on E , and $f = g$ a.e. on E , then g is measurable on E .
 - For a measurable subset $D \subseteq E$, f is measurable on E if and only if f is measurable on D and on $E \setminus D$.
-

Proof. Naturally, we need to separate the cases and define $A = \{x \in E : f(x) \neq g(x)\}$, and then consider A and $E \setminus A$ separately. We have

$$\{x \in E : g(x) > c\} = \{x \in A : g(x) > c\} \cup \{x \in E \setminus A : g(x) > c\}$$

and

$$\{x \in E \setminus A : g(x) > c\} = \{x \in E \setminus A : f(x) > c\} = \{x \in E : f(x) > c\} \cap (E \setminus A)$$

which is measurable. Since A is a zero measure set, $\{x \in A : g(x) > c\}$ being a subset of a zero measure set is also measurable. Hence, $\{x \in E : g(x) > c\}$ is measurable, and g is measurable on E .

For the second part, we assume f is measurable on E . Then we have

$$\{x \in D : f(x) > c\} = \{x \in E : f(x) > c\} \cap D$$

which is measurable, so does $\{x \in E \setminus D : f(x) > c\}$. Conversely, if f is measurable on D and $E \setminus D$, then we have

$$\{x \in E : f(x) > c\} = \{x \in D : f(x) > c\} \cup \{x \in E \setminus D : f(x) > c\}$$

which is measurable, and f is measurable on E . $\square$

The sum $f + g$ of two measurable extended real-valued functions f and g is not well-defined if f and g take the values $+\infty$ and $-\infty$ at the same point.

If f and g are finite a.e. on E , we define $E_0 \subseteq E$ to be the set of points where f and g are both finite. Then E_0 is measurable, as

$$E_0 = \{x \in E : f(x) \in \mathbb{R}\} \cap \{x \in E : g(x) \in \mathbb{R}\}$$

If the restriction of $f + g$ to E_0 is measurable, as $E \setminus E_0$ is a zero measure set, we can apply any extended real-valued function to $E \setminus E_0$ and the resulting function will be measurable on E . This is the sense in which we consider it unambiguous to state that the sum of two measurable functions that are finite a.e. is measurable.

Similar remarks apply to products. The following proposition tells us that standard algebraic operations performed on measurable functions that are finite a.e. again lead to measurable functions.

Theorem 3.1.1: Algebra of Measurable Functions

Let f and g be measurable extended real-valued functions defined on a measurable set E , and they are finite a.e. on E . Then we have

- **Linearity:** $\forall \alpha, \beta \in \mathbb{R}$ the function $\alpha f + \beta g$ is measurable on E .
- **Product:** The function fg is measurable on E .

Proof. It remains to show that αf is measurable for any $\alpha \in \mathbb{R}$. If $\alpha = 0$, then αf is the zero function, which is measurable. If $\alpha > 0$, then a scaling would do. So it suffice to consider the case $\alpha = \beta = 1$. We write

$$\{x \in E : f(x) + g(x) < c\} = \bigcup_{q \in \mathbb{Q}} \{x \in E : f(x) < q\} \cap \{x \in E : g(x) < c - q\} \quad (3.1)$$

which is a countable union of measurable sets, and hence measurable.

For the product, we write

$$fg = \frac{1}{2} [(f + g)^2 - f^2 - g^2] \quad (3.2)$$

so it suffices to show that f^2 is measurable. But we have

$$\{x \in E : f^2(x) < c\} = \{x \in E : -\sqrt{c} < f(x) < \sqrt{c}\} \quad (3.3)$$

which is measurable, and we are done. $\square$

Many of the properties of functions considered in elementary analysis, including continuity and differentiability, are preserved under the operation of composition of functions. However, the composition of measurable functions may not be measurable.

Example: Non-measurable Compositions

There are two measurable real-valued functions, each defined on all of $\mathbb{R}$, whose composition fails to be measurable.

In the previous chapter, we constructed a continuous, strictly increasing function ψ on $[0, 1]$ and a measurable subset $A \subseteq [0, 1]$ such that $\psi(A)$ is not measurable. Extend ψ to a continuous, strictly increasing function on all of $\mathbb{R}$, and ψ^{-1} is also continuous and strictly increasing on all of $\mathbb{R}$, thus is measurable.

On the other hand, the characteristic function χ_A of A is measurable. Then the composition $\chi_A \circ \psi^{-1}$ is not measurable, since $\chi_A \circ \psi^{-1}(x) \geq 1$ if and only if $x \in \psi(A)$, which is not measurable.

Despite the setback imposed by this example, there is the following useful proposition regarding the preservation of measurability under composition.

Proposition:

Let g be a measurable real-valued function defined on a measurable set E , and let f be a continuous real-valued function defined on all of $\mathbb{R}$. Then the composition $f \circ g$ is measurable on E .

Proof. Quite obvious as the preimage of an open set of a continuous function is open, and the preimage of an open set of a measurable function is measurable. $\square$

Remark:

Why Composition Fails?

The root of the failure in the composition of measurable functions lies in the difference between **Borel** and **Lebesgue** measurability.

- If f were **Borel measurable**, then the preimage of *any* Borel set would be Borel. Since every continuous function g pulls back open sets to open sets (and thus Borel to Borel), the composition of two Borel functions is always Borel.
- However, for **Lebesgue measurable** functions, we only guarantee that the preimage of an **open** set is measurable. We do **not** guarantee that the preimage of a Lebesgue measurable set is measurable.

In the example $\chi_A \circ \psi^{-1}$, the set A is Lebesgue measurable, but its preimage under the continuous function ψ^{-1} is $\psi(A)$, which is non-measurable. The continuous function “distorted” the measure-zero set A into a non-measurable “monster” $\psi(A)$.

An immediate important consequence of the above composition result is that if f is a measurable function on a measurable set E , then $|f|$ is also measurable, and every $|f|^p$ is measurable for $p > 0$. Finite maxima and minima of measurable functions are also measurable, we denote

$$\max\{f, g\}(x) = \max\{f(x), g(x)\}, \quad \min\{f, g\}(x) = \min\{f(x), g(x)\}$$

Proposition: **Finite Extrema Functions**

For a finite family of measurable functions $\{f_1, f_2, \dots, f_n\}$ defined on a measurable set E , the functions $\max\{f_1, f_2, \dots, f_n\}$ and $\min\{f_1, f_2, \dots, f_n\}$ are measurable on E .

Proof. We have

$$\{x \in E : \max\{f_1, f_2, \dots, f_n\}(x) > c\} = \bigcup_{i=1}^n \{x \in E : f_i(x) > c\}$$

which is a finite union of measurable sets, and hence measurable. The proof for the minimum is similar. $\square$

We also denote

$$|f|(x) = \max\{f(x), -f(x)\}, \quad f^+(x) = \max\{f(x), 0\}, \quad f^-(x) = \max\{-f(x), 0\}$$

for convenience, being the absolute value, positive part, and negative part of f , respectively. Then we have $f = f^+ - f^-$ and $|f| = f^+ + f^-$.

3.2 Sequential Pointwise Limits and Simple Approximations

For a sequence of functions $\{f_n\}$ defined on some set E and a function f defined on E , there are several ways to define what it means for $\{f_n\}$ to converge to f on E . In this chapter we consider the concepts of pointwise convergence and uniform convergence, which are familiar from elementary analysis. In later chapters we consider many other modes of convergence for a sequence of functions.

Definition 3.2.1: Convergence

For a sequence of functions $\{f_n\}$ defined on some set E and a function f defined on E , and $A \subseteq E$, we say that

- $\{f_n\}$ converges to f **pointwise** on A if for every $x \in A$, $\lim_{n \rightarrow \infty} f_n(x) = f(x)$.
- $\{f_n\}$ converges to f **a.e. pointwise** on A if there exists a zero measure set $N \subseteq A$ such that $\{f_n\}$ converges to f pointwise on $A \setminus N$.
- $\{f_n\}$ converges to f **uniformly** on A if for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$ and all $x \in A$, $|f_n(x) - f(x)| < \epsilon$.

The pointwise limit of continuous functions may not be continuous. The pointwise limit of Riemann integrable functions may not be Riemann integrable. The following proposition is the first indication that the measurable functions have much better stability properties.

Proposition:

Let $\{f_n\}$ be a sequence of measurable functions on E that converges pointwise a.e. on E to a function f , then f is measurable.

Proof. We can just assume that $\{f_n\}$ converges pointwise on E to f , otherwise replace E with the subset of E where the convergence occurs, and the remaining set is a zero measure set.

To proceed, we note that for some x , $f(x) < c$ if and only if there exists a rational number $q < c$ such that $f(x) \leq q$, if and only if there exists a rational number $q < c$ and an $N \in \mathbb{N}$ such that $f_n(x) \leq q$ for all $n \geq N$. Thus we have

$$\{x \in E : f(x) < c\} = \bigcup_{q \in \mathbb{Q}, q < c} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \{x \in E : f_n(x) \leq q\}$$

which is a countable union of measurable sets, and hence measurable. Therefore, f is measurable. $\square$

Just like the step function being the building block of the Riemann integral, the simple function is the building block of the Lebesgue integral. A simple function is a linear combination of characteristic functions of measurable sets.

Definition 3.2.2: Simple Functions

A real-valued function φ defined on a measurable set E is called a **simple function** if it is measurable and takes only a finite number of values.

in other words, a simple function can be expressed as

$$\varphi(x) = \sum_{k=1}^n c_k \chi_{E_k}(x), \quad E_k = \{x \in E : \varphi(x) = c_k\} \quad (3.4)$$

being a finite linear combination of characteristic functions of measurable sets E_k . This expression is called **the canonical representation** of φ .

Simple functions are important because they are easy to understand and manipulate, and they can be used to approximate more complex measurable functions. In Riemann integration, we partition the domain into subintervals and approximate the function by a step function that takes constant values on each subinterval. In Lebesgue integration, we partition the range of the function into intervals and approximate the function by a simple function that takes constant values on the preimages of these intervals.

Lemma 3.2.1: The Simple Approximation Lemma

Let f be a measurable real-valued function defined on a measurable set E . Assume f is bounded on E . Then for every $\epsilon > 0$, there exists simple functions φ_ϵ and ψ_ϵ defined on E such that

$$\varphi_\epsilon(x) \leq f(x) \leq \psi_\epsilon(x), \quad 0 \leq \psi_\epsilon(x) - \varphi_\epsilon(x) < \epsilon, \quad \forall x \in E \quad (3.5)$$

Proof. Let (c, d) be an open interval containing $f(E)$, and we apply a partition

$$c = y_0 < y_1 < \dots < y_n = d$$

where $y_k - y_{k-1} < \epsilon$ for all k . Then we define $I_k = [y_{k-1}, y_k)$ for $k = 1, 2, \dots, n-1$ and $E_k = f^{-1}(I_k)$, which is measurable as f is measurable. Then we define

$$\varphi_\epsilon(x) = \sum_{k=1}^n y_{k-1} \chi_{E_k}(x), \quad \psi_\epsilon(x) = \sum_{k=1}^n y_k \chi_{E_k}(x) \quad (3.6)$$

which is the desired simple functions. □

Theorem 3.2.1: The Simple Approximation Theorem

An extended real-valued function f on a measurable set E is measurable if and only if there exists a sequence of simple functions $\{\varphi_n\}$ defined on E such that it converges to f pointwise on E and

$$|\varphi_n(x)| \leq |f(x)|, \quad \forall x \in E, n \in \mathbb{N} \quad (3.7)$$

If f is nonnegative, then we may choose $\{\varphi_n\}$ to be increasing.

Proof. If f is the limit of a sequence of simple functions $\{\varphi_n\}$, then f is measurable as simple functions are measurable and the pointwise limit of measurable functions is measurable.

Conversely, if f is measurable, we first assume that $f \geq 0$. To handle the possibility that f may be unbounded, we define $E_n = \{x \in E : f(x) < n\}$, which is measurable as f is measurable. Then we apply the Simple Approximation Lemma to f on E_n with $\epsilon = 1/2^n$, and we obtain simple functions φ_n and ψ_n defined on E_n such that

$$\varphi_n(x) \leq f(x) \leq \psi_n(x), \quad 0 \leq \psi_n(x) - \varphi_n(x) < 1/2^n, \quad \forall x \in E_n \quad (3.8)$$

Extend φ_n to E by defining $\varphi_n(x) = n$ for $x \in E \setminus E_n$. Then φ_n is a simple function on E , and we have $0 \leq f(x) - \varphi_n(x) < 1/n$ for $x \in E_n$. It is very easy to see that φ_n converges to f pointwise on E : if $f(x)$ is finite, just take sufficiently large n such that $x \in E_n$, and if $f(x) = +\infty$, then $\varphi_n(x) = n$ for all sufficiently large n , and hence $\varphi_n(x) \rightarrow +\infty = f(x)$.

To make it increasing, we add a requirement that the partition of the interval $[0, n]$ is evenly spaced with interval length $1/2^n$, and hence is a refinement of the partition of the previous step. Then we have $\varphi_n \leq \varphi_{n+1}$ for all n .

For the general case, we can write $f = f^+ - f^-$, and apply the above argument to f^+ and f^- separately. $\square$

3.3 Littlewood's Three Principles, Egoroff's Theorem, and Luzin's Theorem

Speaking of the theory of functions of a real variable, J. E. Littlewood says, “The extent of knowledge required is nothing like so great as is sometimes supposed. There are three principles, roughly expressible in the following terms: Every [measurable] set is nearly a finite union of intervals; every [measurable] function is nearly continuous; every pointwise convergent sequence of [measurable] functions is nearly uniformly convergent. Most of the results of [the theory] are fairly intuitive applications of these ideas, and the student armed with them should be equal to most occasions when real variable theory is called for. If one of the principles would be the obvious means to settle the problem if it were ‘quite’ true, it is natural to ask if the ‘nearly’ is near enough, and for a problem that is actually solvable it generally is.”

We have already seen a formal formulation of Littlewood's first principle, 2.4.2, that a measurable set with finite measure E , we know that for any $\epsilon > 0$, there is a finite disjoint collection of open intervals, whose union $\mathcal{U}$ is “nearly” equal to E in the sense that $m(E \setminus \mathcal{U}) + m(\mathcal{U} \setminus E) < \epsilon$.

A precise realization of the last of Littlewood's principle is the following surprising theorem.

Theorem 3.3.1: Egoroff's Theorem

Assume that E has finite measure, and let $\{f_n\}$ be a sequence of measurable functions on E that converges pointwise on E to a real-valued function f . Then for every $\epsilon > 0$, there exists a closed set $F \subseteq E$ such that $m(E \setminus F) < \epsilon$ and $\{f_n\}$ converges to f uniformly on F .

Remark:

We have already done this frequently in basic analysis, where most questions require the usage of “inner compact uniform convergence” with dealing with problematic phenomena at boundaries. For example, take x^n on $[0, 1]$, which converges pointwise to the function

$f(x) = 0$ for $x \in [0, 1)$ and $f(1) = 1$. This convergence is not uniform on $[0, 1]$, but it is uniform on any closed interval $[0, 1 - \delta]$ for $\delta > 0$. This is a special case of Egoroff's theorem.

To prove it, we first establish a lemma

Proposition:

Under the assumptions of Egoroff's Theorem, for each $\eta > 0$ and $\delta > 0$, there exists a measurable subset $A \subseteq E$ and index N such that $m(E \setminus A) < \delta$ and $|f_n(x) - f(x)| < \eta$ for all $x \in A$ and $n \geq N$.

Remark:

It may seem identical to the conclusion of Egoroff's theorem, but it is not. The difference is that the set A may depend on η and δ , while in Egoroff's theorem, the set F is independent of ϵ .

Proof. Just take

$$E_n = \{x \in E : |f_k(x) - f(x)| < \eta, \forall k \geq n\} = \bigcap_{k=n}^{\infty} \{x \in E : |f_k(x) - f(x)| < \eta\}$$

which is measurable as f_n and f are measurable. Then E_n is increasing and $\bigcup_{n=1}^{\infty} E_n = E$, so we can take N sufficiently large such that $m(E \setminus E_N) < \delta$, and then take $A = E_N$. $\square$

Proof of Egoroff's Theorem.

Proof. The Lemma gives us a set where the sequence is “nearly uniform” for one fixed η . But for uniform convergence, you need the sequence to be “nearly uniform” for all η at the same time. We can achieve this by applying the Lemma infinitely many times for smaller and smaller η , and then intersecting the results.

Take $\eta_n = 1/n$ and $\delta_n = \epsilon/2^n$, and apply the Lemma to get a measurable set $A_n \subseteq E$ and index N_n such that $m(E \setminus A_n) < \delta_n$ and $|f_k(x) - f(x)| < \eta_n$ for all $x \in A_n$ and $k \geq N_n$. Let $A = \bigcap_{n=1}^{\infty} A_n$. Then A is measurable and

$$m(E \setminus A) \leq \sum_{n=1}^{\infty} m(E \setminus A_n) < \sum_{n=1}^{\infty} \delta_n = \epsilon$$

Now, for any $\eta > 0$, choose sufficiently large n such that $\eta_n < \eta$. Then for all $x \in A \subseteq A_n$ and $k \geq N_n$, we have $|f_k(x) - f(x)| < \eta_n < \eta$. This shows that $\{f_n\}$ converges to f uniformly on A .

If we further require A to be closed, we can use the inner approximation property of measurable sets, just take a closed set $F \subseteq A$ such that $m(A \setminus F) < \epsilon$, then $m(E \setminus F) \leq m(E \setminus A) + m(A \setminus F) < 2\epsilon$. We can replace ϵ with $\epsilon/2$ in the above argument to get the desired result. $\square$

Remark:

It is obvious that Egoroff's theorem also holds when the convergence is pointwise a.e. and the limit function is finite a.e. This is because we can just replace E with the subset of E where the convergence and finiteness occur, and the remaining set is a zero measure set.

We now present a precise version of Littlewood's second principle in the case the measurable function is simple and then use this special case to prove the general case of the principle, Lusin's Theorem.

Proposition:

Let f be a simple function defined on a measurable set E . Then for every $\epsilon > 0$, there exists a continuous function g defined on $\mathbb{R}$ and a closed set $F \subseteq E$ such that $m(E \setminus F) < \epsilon$ and $f(x) = g(x)$ for all $x \in F$.

Proof. We write the canonical representation of f as

$$f(x) = \sum_{k=1}^n a_k \chi_{E_k}(x), \quad E_k = \{x \in E : f(x) = a_k\}$$

In each E_k , we can find a closed set $F_k \subseteq E_k$ such that $m(E_k \setminus F_k) < \epsilon/n$. Let $F = \bigcup_{k=1}^n F_k$, then F is closed and $m(E \setminus F) < \epsilon$. Now we can define a continuous function g on $\mathbb{R}$ such that $g(x) = f(x)$ for all $x \in F$.

To extend g to a continuous function on $\mathbb{R}$, consider $\mathbb{R} \setminus F$, which is open, can be expressed as a countable union of disjoint open intervals. On each of these intervals, we can define g to be a linear interpolation between the values of f on the endpoints of the interval (if they exist) or extend it to a constant value if the interval is unbounded. This ensures that g is continuous on $\mathbb{R}$ and agrees with f on the closed set F . $\square$

Remark:

Actually, this need a more careful consideration here. We need to differ the case $f|_E$ is continuous and $f|_{\mathbb{R}}$ on E . The former take the subspace topology of E , and the latter take the standard topology of $\mathbb{R}$. As there may exists open sets in subspace topology that are not open in standard topology, we need to be careful.

From the definitions, the subspace topology on E is defined as taking intersection of open sets in $\mathbb{R}$ with E . So if an open set in $\mathbb{R}$ is already contained in E , then it is also open in the subspace topology. From this, we can say that

- If a function f is continuous on $\mathbb{R}$, then $f|_E$ is continuous on E .
- If a function $f|_E$ is continuous on E , then f may not be continuous on $\mathbb{R}$, take for example, the Dirichlet function and $E = \mathbb{Q}$, then $f|_E$ is continuous on E , but f is nowhere continuous on $\mathbb{R}$.

Here, the limit of continuous uniformly convergent functions is continuous only gives the continuity of the limit function $g|_E$ is continuous on E , and actually we need to prove that g is continuous on E . A general result here is called the **Tietze Extension Theorem**.

A simplified proof is available for the case $X = \mathbb{R}$.

Proposition:

Suppose f is a continuous function defined on a closed set $F \subseteq \mathbb{R}$. Then there exists a continuous function g defined on $\mathbb{R}$ such that $g(x) = f(x)$ for all $x \in F$.

Proof. We have already constructed a function g on $\mathbb{R}$ that agrees with f on F and is continuous on $\mathbb{R} \setminus F$. To show that g is continuous on all of $\mathbb{R}$, we need to check the continuity at points in F .

Take $x_0 \in F$, we know that $g|_F$ is continuous at x_0 . We take any sequence of points $\{x_n\}$ in $\mathbb{R}$ that converges to x_0 . And, take a subsequence that have one of the following cases, we try from the first case, and if one fails, we try the next one:

- If the subsequence is contained in F , then we're done, as $g|_F$ is continuous at x_0 .
- If we cannot find a subsequence contained in F , then it seems that later in the sequence, all points are in $\mathbb{R} \setminus F$. Now, if we can find a subsequence contained in finitely many intervals of $\mathbb{R} \setminus F$, then x_0 must be an endpoint of one of these intervals, and a subsequence of this subsequence is contained in the interval, and we can use the interpolation to conclude that $g(x_n) \rightarrow g(x_0)$.
- If we cannot find a subsequence contained in finitely many intervals of $\mathbb{R} \setminus F$, then any neighborhood of x_0 contains infinitely many intervals of $\mathbb{R} \setminus F$. We find a subsequence that has one point in each of these intervals, and find a subsequence of this subsequence that is monotonically increasing or decreasing. Then, take two other sequences of points in F : one sequence is the bigger endpoint of the interval that contains the point in the subsequence, and the other sequence is the smaller endpoint of the interval that contains the point in the subsequence. Both of these sequences converge to x_0 , and by the continuity of $g|_F$, we have g evaluated at these sequences converges to $g(x_0)$. By the squeeze theorem, we conclude that $g(x_n) \rightarrow g(x_0)$.

□

Theorem 3.3.2: Luzin's Theorem

Let f be a real-valued measurable function defined on a measurable set E . Then for every $\epsilon > 0$, there exists a continuous function g defined on $\mathbb{R}$ and a closed set $F \subseteq E$ such that $m(E \setminus F) < \epsilon$ and $f(x) = g(x)$ for all $x \in F$.

Proof. First, assume that $m(E) < \infty$. By the Simple Approximation Theorem ??, there exists a sequence of simple functions $\{\varphi_n\}$ defined on E such that $\varphi_n \rightarrow f$ pointwise on E .

Apply the previous proposition to each simple function φ_n with $\epsilon/2^n$ to obtain a continuous function g_n and a closed set $F_n \subseteq E$ such that $m(E \setminus F_n) < \epsilon/2^{n+1}$ and $\varphi_n(x) = g_n(x)$ for all $x \in F_n$.

As we need to ensure that the limit of $\{g_n\}$ converges to a continuous function, we need uniform convergence. By the Egoroff's Theorem, we can find a closed set $F_0 \subseteq E$ such that $m(E \setminus F_0) < \epsilon/2$ and $\varphi_n \rightarrow f$ uniformly on F_0 . This time, we can take $F = F_0 \cap \bigcap_{n=1}^{\infty} F_n$, which is closed and satisfies $m(E \setminus F) < \epsilon$.

On F , we have $g_n = \varphi_n \rightarrow f$ uniformly, and since each g_n is continuous, the uniform limit of continuous functions is continuous. Therefore, we can define $g = \lim_{n \rightarrow \infty} g_n$ on F , which is continuous on F and agrees with f on F . An extension of g to all of $\mathbb{R}$ can be done similarly as in the previous proposition, ensuring that g is continuous on $\mathbb{R}$. (The finite measure assumption is crucial here to ensure the validity of Egoroff's theorem.)

The Infinite Measure Case ($m(E) = \infty$) We partition the real line into a countable collection of disjoint, bounded intervals. For each $n \in \mathbb{Z}$, let $I_n = [n, n+1)$ and define $E_n = E \cap I_n$. Each E_n is a measurable set of finite measure ($m(E_n) \leq 1$).

Applying the result to each E_n , for a given $\epsilon > 0$, there exists a closed set $F_n \subseteq E_n$ such that:

$$m(E_n \setminus F_n) < \frac{\epsilon}{2^{|n|+2}} \quad \text{and} \quad f|_{F_n} \text{ is continuous.}$$

Define $F = \bigcup_{n \in \mathbb{Z}} F_n$. Since $F_n \subseteq E_n$, it follows that $F \subseteq E$. The measure of the excluded set is:

$$m(E \setminus F) = m\left(\bigcup_{n \in \mathbb{Z}} (E_n \setminus F_n)\right) = \sum_{n=-\infty}^{\infty} m(E_n \setminus F_n) < \sum_{n=-\infty}^{\infty} \frac{\epsilon}{2^{|n|+2}} = \epsilon.$$

We must show that F is closed. Let $\{x_k\}$ be a sequence in F such that $x_k \rightarrow x$. Since the sequence converges, it is bounded, meaning there exists some $M > 0$ such that $\{x_k\} \subseteq [-M, M]$. Consequently, the sequence $\{x_k\}$ can only intersect a finite number of the sets $\{F_n\}$. Because the union of a finite collection of closed sets is closed, and each F_n is closed, the limit x must belong to $\bigcup_{n=-M-1}^{M+1} F_n \subseteq F$. Thus, F is closed.

Furthermore, $f|_F$ is continuous. For any $x \in F$, x belongs to some F_n . In a sufficiently small neighborhood of x , f only interacts with finite many F_k . Since f is continuous on each of these closed sets, $f|_F$ is continuous at x .

Finally, by the previously established extension proposition, there exists a continuous function g defined on all of $\mathbb{R}$ such that $g(x) = f(x)$ for all $x \in F$. This completes the proof. $\square$

Chapter 4

Lebesgue Integration

Lebesgue integration is a fundamental concept in real analysis that extends the notion of integration beyond the limitations of Riemann integration. It allows for the integration of a wider class of functions and provides a more robust framework for analyzing convergence and measure theory.

4.1 The Riemann Integral

We begin by recalling the definition and properties of the Riemann integral.

Let f be a bounded real-valued function defined on a closed interval $[a, b]$. Let $P = \{x_0, \dots, x_n\}$ be a partition of the interval $[a, b]$, given by

$$a = x_0 < x_1 < \dots < x_n = b,$$

Now define the lower and upper **Darboux sums** of f with respect to the partition P as follows:

$$L(f, P) = \sum_{i=1}^n m_i(x_i - x_{i-1}) \quad \text{and} \quad U(f, P) = \sum_{i=1}^n M_i(x_i - x_{i-1}), \quad (4.1)$$

where

$$m_i = \inf\{f(x) : x \in (x_{i-1}, x_i)\} \quad \text{and} \quad M_i = \sup\{f(x) : x \in (x_{i-1}, x_i)\}. \quad (4.2)$$

then we define the **lower and upper Riemann integrals** of f on $[a, b]$ as

$$\begin{aligned} (R) \int_a^b f &= \sup\{L(f, P) : P \text{ is a partition of } [a, b]\} \\ \underline{\int_a^b} f &= \inf\{U(f, P) : P \text{ is a partition of } [a, b]\}. \end{aligned} \quad (4.3)$$

Since f is assumed bounded so the lower and upper Riemann integrals are finite. The upper integral is always at least as large as the lower integral, and if the two are equal we say that f is **Riemann integrable** on $[a, b]$, and we denote the common value by

$$(R) \int_a^b f = \underline{\int_a^b} f = \overline{\int_a^b} f. \quad (4.4)$$

to temporarily distinguish it from the Lebesgue integral, which we consider in the next section.

A real-valued function ψ is called a **step function** on $[a, b]$ if there exists a partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$ such that ψ is constant c_i on each open subinterval (x_{i-1}, x_i) for $i = 1, \dots, n$. Then naturally

$$L(\psi, P) = \sum_{i=1}^n c_i(x_i - x_{i-1}) = U(\psi, P),$$

so the step function ψ is Riemann integrable and we have

$$(R) \int_a^b \psi = \sum_{i=1}^n c_i(x_i - x_{i-1}).$$

Therefore, we may reformulate the definition of the lower and upper Riemann integrals as follows:

$$\begin{aligned} (R) \int_a^b f &= \sup \left\{ (R) \int_a^b \psi : \psi \text{ is a step function on } [a, b] \text{ and } \psi \leq f \right\} \\ \overline{(R) \int_a^b f} &= \inf \left\{ (R) \int_a^b \psi : \psi \text{ is a step function on } [a, b] \text{ and } f \leq \psi \right\}. \end{aligned} \tag{4.5}$$

Example: **Dirichlet Function**

The Dirichlet function $D : [0, 1] \rightarrow \mathbb{R}$ is defined by

$$D(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \cap [0, 1], \\ 0, & \text{if } x \in [0, 1] \setminus \mathbb{Q}. \end{cases}$$

We easily see that D is bounded, but it is not Riemann integrable on $[0, 1]$, for $L(D, P) = 0$ and $U(D, P) = 1$ for any partition P of $[0, 1]$.

As rationals are countable, we can find a sequence of Riemann integrable functions $\{f_n\}$ that converges pointwise to D on $[0, 1]$, by just defining f_n to be the function that is equal to 1 on the first n rational numbers in $[0, 1]$ and equal to 0 elsewhere. Each f_n is Riemann integrable, but the limit function D is not.

Remark:

Step Functions vs. Simple Functions

The reformulation of the Riemann integral using step functions is the key to understanding the Lebesgue integral.

- **Riemann Approach:** Approximates f using **step functions** (constant on intervals). This partitions the *domain* into chunks of length Δx .
- **Lebesgue Approach:** Approximates f using **simple functions** (constant on measurable sets). This effectively partitions the *range* into chunks of height Δy , where the “width” of the chunk is given by the measure $m(E_i)$.

Since the collection of measurable sets is much richer than the collection of intervals, simple functions can “track” the behavior of functions like the Dirichlet function where step functions fail.

4.2 Lebesgue Integral: Bounded Functions, Finite Measure

The Dirichlet function, which was examined in the preceding section, exhibits one of the principal shortcomings of the Riemann integral: a uniformly bounded sequence of Riemann integrable functions on a closed, bounded interval can converge pointwise to a function that is not Riemann integrable. We will see that the Lebesgue integral does not suffer from this shortcoming.

Henceforth we only consider the Lebesgue integral, unless explicitly mentioned otherwise, and so we use the pure integral symbol to denote the Lebesgue integral.

Definition 4.2.1: Lebesgue Integral of Simple Functions

Let $E \subseteq \mathbb{R}$ be a measurable set with finite measure, and let ϕ be a simple function on E given by

$$\phi = \sum_{i=1}^n a_i \chi_{E_i},$$

where $a_i \in \mathbb{R}$ and E_i are measurable subsets of E . The Lebesgue integral of ϕ over E is defined as

$$\int_E \phi = \sum_{i=1}^n a_i m(E_i), \quad (4.6)$$

From the additivity of measure, this form holds also when the $\{a_i\}$ are not necessarily distinct, which naturally leads to the addition and multiplication properties of the Lebesgue integral for simple functions.

Proposition: Linearly and Monotonicity of Integration

Let φ and ψ be simple functions on a measurable set E with finite measure, and let $\alpha, \beta \in \mathbb{R}$. Then

$$\int_E (\alpha\varphi + \beta\psi) = \alpha \int_E \varphi + \beta \int_E \psi, \quad (4.7)$$

Moreover, if $\varphi \leq \psi$ on E , then

$$\int_E \varphi \leq \int_E \psi. \quad (4.8)$$

Proof. As there are only finitely many measurable sets in which φ and ψ are both constant, we can write

$$\int_E (\alpha\varphi + \beta\psi) = \sum_{i=1}^n (\alpha a_i + \beta b_i) m(E_i) = \alpha \sum_{i=1}^n a_i m(E_i) + \beta \sum_{i=1}^n b_i m(E_i) = \alpha \int_E \varphi + \beta \int_E \psi.$$

the second part follows from subtracting the two integrals and using the fact that nonnegative simple functions have nonnegative integrals. $\square$

This shows

Proposition:

Let $\{E_i\}_{i=1}^n$ be a finite collection of measurable subsets of a set E with finite measure, not necessarily disjoint, and let $a_i \in \mathbb{R}$. Then if $\phi = \sum_{i=1}^n a_i \chi_{E_i}$ we have

$$\int_E \phi = \sum_{i=1}^n a_i m(E_i). \quad (4.9)$$

Proof. This follows from the linearity of the integral and

$$\int_E \chi_{E_i} = m(E_i).$$

□

A step function takes only a finite number of values and each interval is measurable. Thus a step function is simple. Since the measure of a singleton set is zero and the measure of an interval is its length, we infer from the linearity of Lebesgue integration for simple functions defined on sets of finite measure that the Riemann integral over a closed, bounded interval of a step function agrees with the Lebesgue integral.

Now, let f be a bounded real-valued function defined on a measurable set E with finite measure. By analogy with the Riemann integral, we define the **lower and upper Lebesgue integrals** of f on E as

$$\begin{aligned} \underline{\int_E} f &= \sup \left\{ \int_E \phi : \phi \text{ is a simple function on } E \text{ and } \phi \leq f \right\} \\ \overline{\int_E} f &= \inf \left\{ \int_E \phi : \phi \text{ is a simple function on } E \text{ and } f \leq \phi \right\}. \end{aligned} \quad (4.10)$$

Since f is assumed to be bounded, by the monotonicity property of the integral for simple functions, the lower and upper integrals are finite and the upper integral is always at least as large as the lower integral.

Definition 4.2.2: Lebesgue Integral of Bounded Functions on Finite Measure Sets

A bounded real-valued function f defined on a measurable set E with finite measure is said to be **Lebesgue integrable** on E if the lower and upper Lebesgue integrals of f on E are equal. In this case, we denote the common value by $\int_E f$ and call it the **Lebesgue integral** of f over E .

Theorem 4.2.1: Riemann Integrable and Lebesgue Integrable: Bounded Functions on Finite Measure Sets

Let f be a bounded real-valued function defined on a closed, bounded interval $[a, b]$. Then if f is Riemann integrable on $[a, b]$, it is also Lebesgue integrable on $[a, b]$, and the two integrals are equal.

Proof. As step functions are simple, and for them the Riemann and Lebesgue integrals agree, we

have

$$\begin{aligned}
& \sup\{(R) \int_a^b \psi : \psi \text{ is a step function on } [a, b] \text{ and } \psi \leq f\} \\
& \leq \sup\{\int_a^b \phi : \phi \text{ is a simple function on } [a, b] \text{ and } \phi \leq f\} \\
& \leq \inf\{\int_a^b \phi : \phi \text{ is a simple function on } [a, b] \text{ and } f \leq \phi\} \\
& \leq \inf\{(R) \int_a^b \psi : \psi \text{ is a step function on } [a, b] \text{ and } f \leq \psi\}.
\end{aligned}$$

Riemann integrability of f implies that the first and last expressions are equal, which in turn implies that the second and third expressions are equal, and hence that f is Lebesgue integrable on $[a, b]$ and the two integrals are equal. $\square$

Remark:

We can see now, that the set of simple functions is much richer than the set of step functions, and hence the Lebesgue integral is more general than the Riemann integral. In particular, the Dirichlet function is Lebesgue integrable on $[0, 1]$ with integral equal to 0, since it is equal to 0 almost everywhere on $[0, 1]$. It is itself a simple function, as it can be expressed as $D = 1 \cdot \chi_{\mathbb{Q} \cap [0, 1]} + 0 \cdot \chi_{[0, 1] \setminus \mathbb{Q}}$, but not as a step function, since it is not constant on any interval.

Theorem 4.2.2: Measurable Functions and Lebesgue Integrability

Let f be a bounded measurable function defined on a measurable set E with finite measure. Then f is Lebesgue integrable on E .

Proof. According to the Simple Approximation Theorem, there exists simple functions φ_n and ψ_n such that $\varphi_n \leq f \leq \psi_n$ and $0 \leq \psi_n - \varphi_n \leq 1/n$ for all $n \in \mathbb{N}$. Then we have

$$0 \leq \int_E \psi_n - \int_E \varphi_n = \int_E (\psi_n - \varphi_n) \leq \frac{1}{n} m(E),$$

we also have

$$0 \leq \overline{\int_E f} - \underline{\int_E f} \leq \int_E \psi_n - \int_E \varphi_n \leq \frac{1}{n} m(E),$$

and since n is arbitrary, we conclude that $\overline{\int_E f} = \underline{\int_E f}$, and hence f is Lebesgue integrable on E . $\square$

Remark:

This is a good place to review the intuition of the definition of measurable functions. A function f is measurable if the preimage of any interval is a measurable set. This property allows us to approximate f from below and above by simple functions, which are constant on measurable sets.

Lebesgue integration itself is just the act of partitioning the range of a function into small intervals, and then measuring the size of the preimage of each interval. So we need to ensure that the preimage of each interval is measurable, which is exactly what the definition of a

measurable function guarantees.

It turns out that the converse of the preceding theorem is true; a bounded function on a set of finite measure is Lebesgue integrable if and only if it is measurable: we will prove this later. This shows, in particular, that not every bounded function defined on a set of finite measure is Lebesgue integrable. For example, the characteristic function of a nonmeasurable set is bounded but not Lebesgue integrable, since the lower and upper Lebesgue integrals of such a function are the inner and outer measures of the set, which are not equal.

Theorem 4.2.3: Linearity and Monotonicity of Integration)

Let f and g be bounded measurable functions defined on a measurable set E with finite measure, and let $\alpha, \beta \in \mathbb{R}$. Then

$$\int_E (\alpha f + \beta g) = \alpha \int_E f + \beta \int_E g, \quad (4.11)$$

Moreover, if $f \leq g$ on E , then

$$\int_E f \leq \int_E g. \quad (4.12)$$

Proof. As linear combinations of measurable functions are measurable, $\alpha f + \beta g$ is measurable and thus Lebesgue integrable on E .

For the scalar multiplication, we divide cases. If $\alpha = 0$, then the result is trivial. If $\alpha > 0$, then we have

$$\int_E (\alpha f) = \inf_{\psi \geq \alpha f} \int_E \psi = \alpha \inf_{\psi/\alpha \geq f} \int_E \frac{\psi}{\alpha} = \alpha \int_E f.$$

and the same goes for $\alpha < 0$. So we only need to consider $\alpha = \beta = 1$. Let ψ_1 and ψ_2 be simple functions such that $f \leq \psi_1$ and $g \leq \psi_2$. Then we have $\psi_1 + \psi_2$ is a simple function and $f + g \leq \psi_1 + \psi_2$, so we have

$$\int_E (f + g) \leq \int_E (\psi_1 + \psi_2) = \int_E \psi_1 + \int_E \psi_2.$$

Taking the infimum over all such ψ_1 and ψ_2 , we obtain

$$\int_E (f + g) \leq \inf_{\psi_1 \geq f} \int_E \psi_1 + \inf_{\psi_2 \geq g} \int_E \psi_2 = \int_E f + \int_E g.$$

the next direction follows from the same argument, using the lower bounds of the integrals would do.

The monotonicity property follows from subtracting the two integrals and using the fact that nonnegative measurable functions have nonnegative integrals, as 0 is a lower approximate simple function having integral equal to 0. $\square$

Proposition: Patching Domains

Let f be a bounded measurable function defined on a measurable set E with finite measure,

and let A and B be measurable subsets of E and disjoint. Then

$$\int_{A \cup B} f = \int_A f + \int_B f. \quad (4.13)$$

Proof. Integrating by

$$f \cdot \chi_{A \cup B} = f \cdot \chi_A + f \cdot \chi_B,$$

would do. □

Proposition: **Absolute Value Inequality**

Let f be a bounded measurable function defined on a measurable set E with finite measure. Then

$$\left| \int_E f \right| \leq \int_E |f|. \quad (4.14)$$

Proof. As $|f|$ is measurable and bounded, we have $-|f| \leq f \leq |f|$, and hence by the monotonicity property of the integral, we have our result. □

Proposition: **Uniform Limits**

Let $\{f_n\}$ be a sequence of bounded measurable functions defined on a measurable set E with finite measure, and suppose that f_n converges uniformly to a function f on E . Then f is bounded and measurable on E , and

$$\lim_{n \rightarrow \infty} \int_E f_n = \int_E f. \quad (4.15)$$

Proof. The proof is the same as the proof of the corresponding result for Riemann integrals, just take $\epsilon > 0$ and choose N such that $|f_n(x) - f(x)| < \epsilon/m(E)$ for all $x \in E$ and $n \geq N$. Then we have

$$\left| \int_E f_n - \int_E f \right| = \left| \int_E (f_n - f) \right| \leq \int_E |f_n - f| < \epsilon,$$

so we have $\lim_{n \rightarrow \infty} \int_E f_n = \int_E f$. □

This proposition is rather weak since frequently a sequence will be presented that converges pointwise but not uniformly. It is important to understand when it is possible to infer from pointwise convergence of a sequence of functions to the convergence of the corresponding sequence of integrals. We refer to this equality as **passing the limit under the integral sign**.

We shall illustrate this idea with an example.

Example:

We define a sequence of functions $\{f_n\}$ on $[0, 1]$ by:

- $f_n(x) = 0$ for $x \geq 2/n$,

- $f_n(0) = 0$, $f(1/n) = n$, and f_n is linear on $[0, 1/n]$ and $[1/n, 2/n]$.

which is a sharp triangular spike of height n and base $2/n$. Then we have $\int_0^1 f_n = 1$ for all n , but $f_n \rightarrow 0$ pointwise on $[0, 1]$.

Thus, pointwise convergence alone is not sufficient to justify passage of the limit under the integral sign.

Theorem 4.2.4: Bounded Convergence Theorem

Let $\{f_n\}$ be a sequence of measurable functions defined on a measurable set E with finite measure, and suppose that $\{f_n\}$ is uniformly pointwise bounded on E , i.e., there exists a constant $M > 0$ such that $|f_n(x)| \leq M$ for all $x \in E$ and all $n \in \mathbb{N}$. Then if f_n converges pointwise to a function f on E , we have

$$\lim_{n \rightarrow \infty} \int_E f_n = \int_E f.$$

Proof. The proof of this theorem furnishes a nice illustration of Littlewood's Third Principle. If the convergence is uniform, we have the easy proof of the preceding proposition. However, Egoroff's Theorem tells us, roughly, that pointwise convergence is "nearly uniform" convergence.

The pointwise limit of a sequence of measurable functions is measurable. Therefore f is measurable and $|f| \leq M$ on E , and we take any measurable set $A \subseteq E$ and write

$$\int_E f_n - \int_E f = \int_A (f_n - f) + \int_{E \setminus A} f_n - \int_{E \setminus A} f.$$

We now apply the Egoroff's Theorem. For any $\epsilon > 0$, there exists a measurable set $A \subseteq E$ such that $m(E \setminus A) < \epsilon/(4M)$ and $f_n \rightarrow f$ uniformly on A . Then there is an N such that

$$\forall n \geq N, \forall x \in A, |f_n(x) - f(x)| < \epsilon/(2m(E)).$$

and adding the integrals we have our result. □

Remark:

We notice that the bound M is essential in the proof. It guarantees that small measure sets do not contribute significantly to the integral, allowing us to control the error in the approximation by Egoroff's theorem. Without this uniform bound, the integrals over small measure sets could be arbitrarily large, and the conclusion of the theorem would not hold.

Remark:

Comparison with the Riemann Integral: Arzelà's Theorem

It is noteworthy that a version of this theorem exists for the Riemann integral, known as **Arzelà's Bounded Convergence Theorem** (1885). However, the Riemann version is significantly less powerful due to two major limitations:

- **The Integrability of the Limit:** In Riemann theory, the pointwise limit of integrable functions is not necessarily integrable (as seen with the sequence $\{f_n\}$ converging to the Dirichlet function D). Therefore, Arzelà's theorem must *assume* as a hypothesis that

the limit function f is Riemann integrable. In Lebesgue theory, we get the measurability of f “for free.”

- **The Difficulty of Proof:** Without the tools of measure theory (specifically Egoroff’s Theorem), Arzelà’s original proof was notoriously difficult and technically complex.

By using Egoroff’s Theorem, the Lebesgue theory effectively “decomposes” the problem: we use measure theory to handle the “bad” sets of points where convergence is slow, and use the uniform bound M to ensure these sets do not contribute significantly to the integral. This illustrates how the Lebesgue integral “repairs” the convergence defects of the Riemann integral.

Prior to the proof of the Bounded Convergence Theorem, no use was made of the countable additivity of Lebesgue measure on the real line. Only finite additivity was used, and it was used just once, in the proof of linearity of simple function integration. But for the proof of the Bounded Convergence Theorem we used Egoroff’s Theorem. The proof of Egoroff’s Theorem needed the continuity of Lebesgue measure, a consequence of countable additivity of Lebesgue measure.

4.3 Lebesgue Integral: Nonnegative Functions

A measurable function f on E is said to vanish **outside a set of finite measure** if there exists a measurable set $E_0 \subseteq E$ with finite measure such that $f = 0$ on $E \setminus E_0$. In measure theory, we often define the **support** of a function f as the set

$$\text{supp}(f) = \{x \in E : f(x) \neq 0\}, \quad (4.16)$$